The *'Everything and Anything'*
About a Project on the Cyclorotor Autogyro Concept

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Preamble

This document contains all the information regarding design processes for mechanical and/or electrical systems and learnings from making mistakes and from trying out various iterations– basically *"Everything and Anything"*. As the project goes on, ideas will change, experiments which were intended to be performed will change, so this document serves as a record for all those changes– the intention is to show the *'evolution'* of this project. This will also help in keeping a concrete understanding of the physical and mathematical concepts we would come across while progressing through the work.

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1 Project Brief

1.1 Why This Project? And Posing the Problem Statement.

My interest lies in mechanical systems- particulary of the propulsions sub-section of aerospace engineering. Initially, I had worked on building a motorized cyclorotor with the aim to develop a quad-rotor cyclocopter. A vehicle like the cyclocopter had really caught my eye- an incredibly intriguing mechanism to say, nonetheless. The most intriguing part of this propeller is the thrust vectoring capability. All blades in a cyclorotor continuosly pitch cyclically. By adjusting the offset link, you are able to direct thrust in different directions.

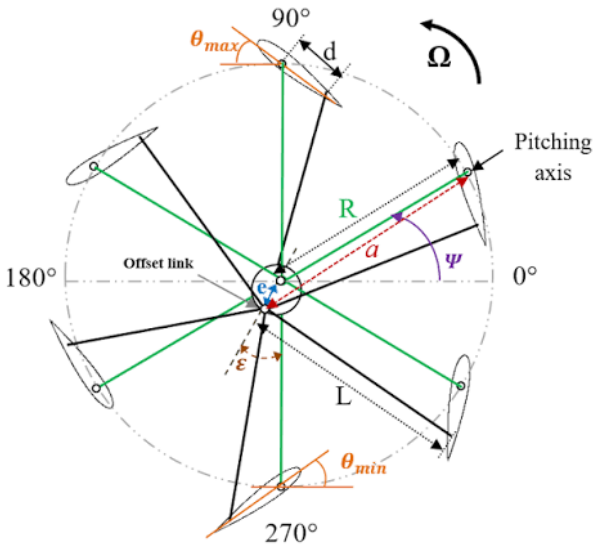


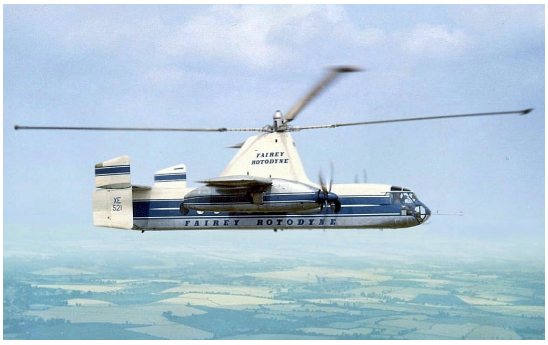
Figure 1: Side view of a cyclorotor. Credit- https://www.mdpi.com/2076-0825/12/12/436

Motivated to work on a thrust vectoring propeller- my aim was not only just to replicate but to learn how such a propeller generates lift, to identify gaps in the literature, possibly reinvent the proppeller itself. It was also to explore some unanswered questions- could the cyclorotor be used as a propulsion device for aerial + underwater services? Or could the propeller be used as a turbine to generate electricity for aircraft (RAT- RAM Air Turbine)?

The 'turbine' aspect of the rotor begs one question- Is it possible to use the cyclorotor as an autogyro? The autogyros we see were inspired by helicopter rotors. An autogyro in short is a freely spinning propeller which genrates lift using free stream velocity.



(a) The Modern Autogyro



(b) The Fairey Rotodyne- A British-made Gyroplane. A helicopter-plane hybrid made for fuel efficient intercity travel.



(c) The Darrius Windmill. Credit- https://en.wind-turbine-models.com/turbines/1569-ratier-figeac-darrieus



(d) The Savonius Windmill. Credit- https://www.rx-energy.com/wind-turbine/vertical-wind-turbine10kw-vertical-wind-turbine-generator-spiral.html

The cyclorotor is based on the Darrius windmill. The Darrius windmill by nature, is not an auto-starting windmill, but the addition of the savonius windmill takes care of this issue. Similar to the helicopter inspiration, can the motorized cyclorotor inspire a new propulsion system to possibly offer more stability and energy efficiency for drones?

1.2 Significance

The Cyclorotor Autogyro concept has been theorised only in a single paper publised in the *International Journal of Heat and Technology* (add ref) under the title *Energetic design of a new autogyro aircraft with cyclorotors with possibility of energy harvesting.*

The paper considers a hypothesized aircraft design- two rotating cylindrical wind turbines placed on the sides of the fuselage.

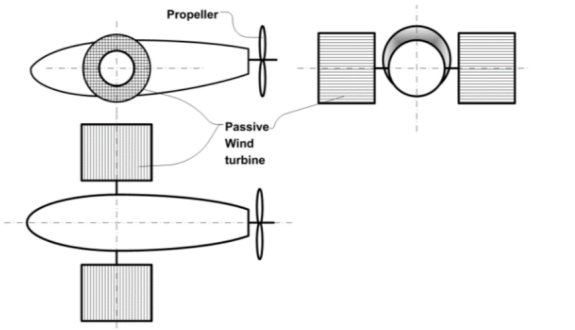


Figure 3: Theorized Cylorotor Autrogyro in the paper.

Using a bunch of math, describing equations for the vehicle and wind turbine, they've calculated various quantities like average lift and drag of a blade, torque, required area of wings, area of one blade, etc. at certain wind speeds. Since no physical prototype has been made to validate these results as of writing this paper (November 2026), the goal of this project is to build an aircraft with Cyclorotor Autogyros.

As a student pursuing Robotics and Autonomous Systems, this project will give a deeper understanding of fluid mechanics, aerodynamics, structural engineering, theorizing and physical intuitions through ideating, prototyping and building.

1.3 Project Breakdown

This Project is broken down into *"n"* stages. The value for *"n"* will be decided based on how much progress is made and how far I would get, but the idea is to have core aspects of this projects segmented to provide a structure and continuity to this project.

2 Stage 1

2.1 Learning CFD

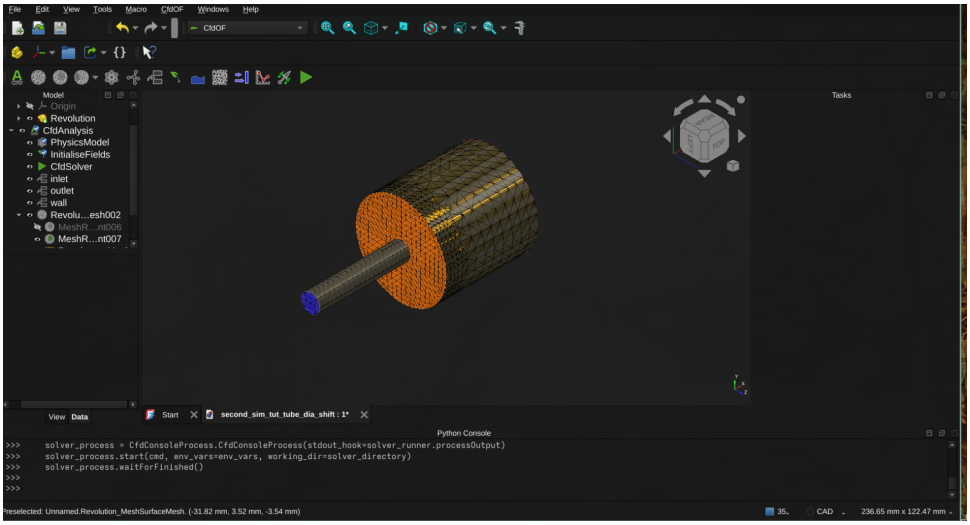
Upon discussion with Professor Shashikant whose area of expertise is fluid mechanics and computational fluid dynamics, he had recommended me to get started with OpenFOAM- an opensource CFD application. OpenFOAM doesn't have a GUI, one has to edit physics settings, mesh cases, graphs, etc. via command terminals. It is incredibly flexible in the sense that one can write their own solvers for simulations. Since, my background was not in programming- especially writing code for physical simulations, I had opted to go for the *FreeCAD cfdof* workbench.

This setup allows me to model a virtual wind tunnel and play around with simulation settings in FreeCAD itself. Meshing can be viewed in-app or with the help of *Paraview*. Paraview offers more versatility when viewing meshes and it is also the platform to visualize the simulations.

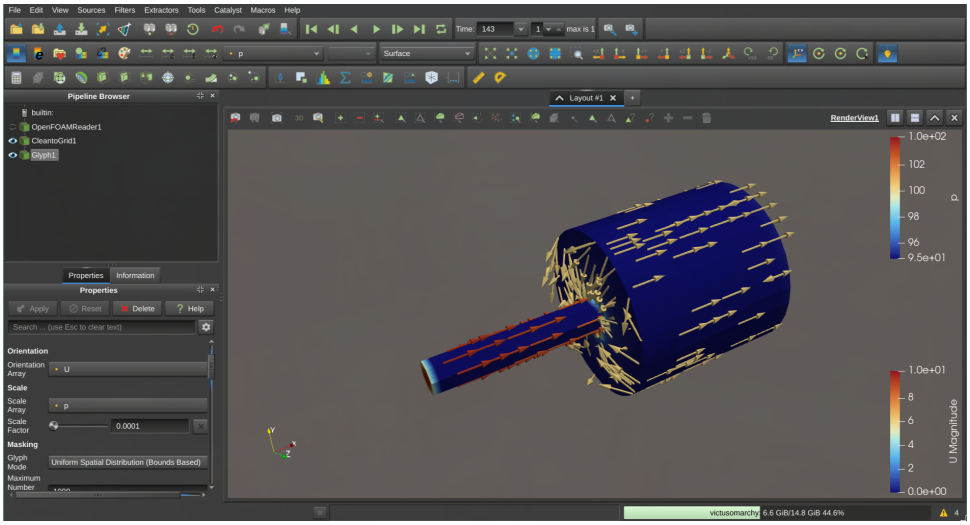
Before building wind tunnel, it was essential to learn the basics of the *cfdof* workbench. Thanks to "Tech Bernd" (Channel link- https://www.youtube.com/@techbernd5535), I got myself used to working with the essentials such as the meshing and refinement (how, when and where to refine in a case). With the help of these tutorials I was able to form a generalized approach to CFD.

These tutorials also offered various visualization techniques using paraview such as the *glyphs* and *streamlines*.

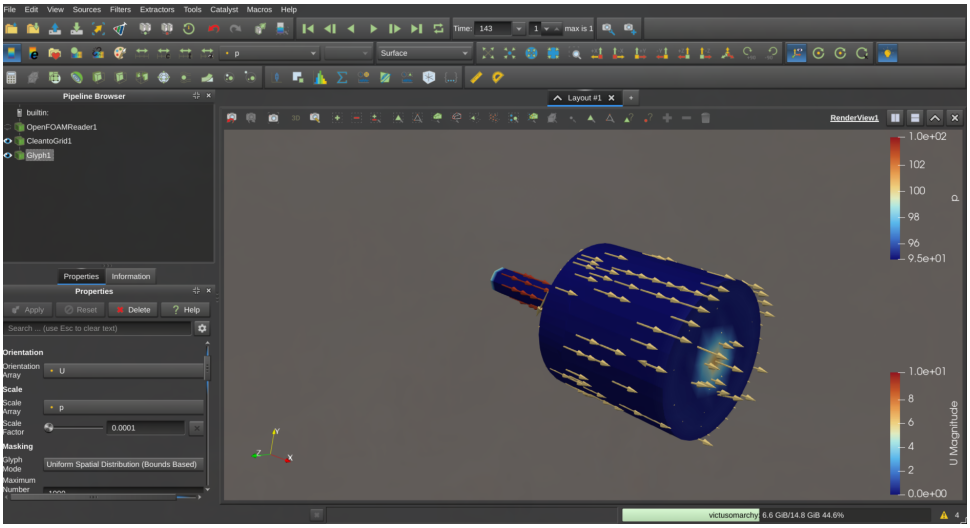
Here are some examples:



(a) Mesh Case in FreeCAD using cfdof workbench

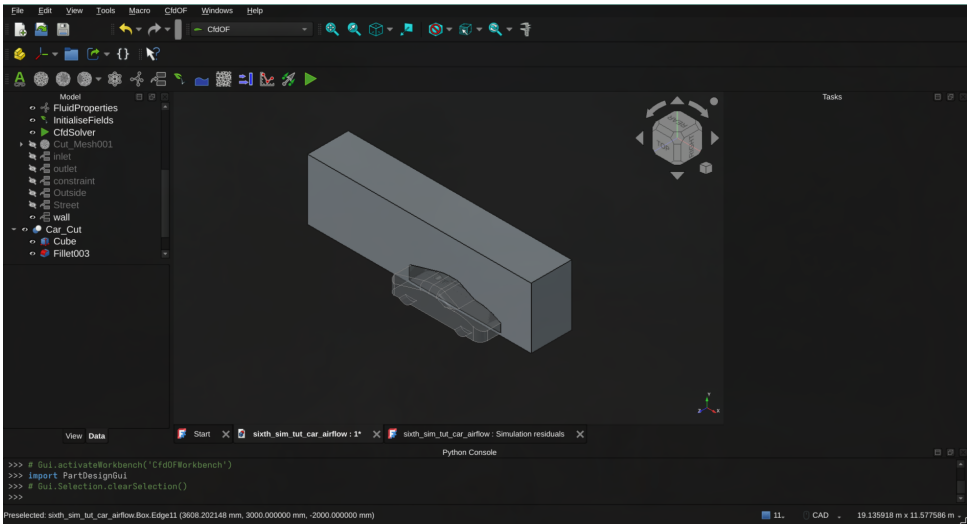


(b) Flow visualization in Paraview using "Glyphs"- in short arrows that point in the direction of the flow

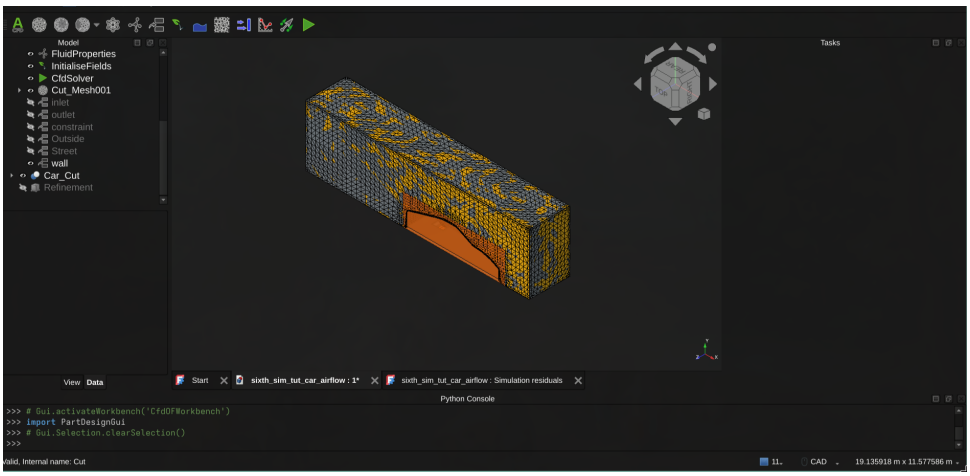


(c) Another isometric view in Paraview

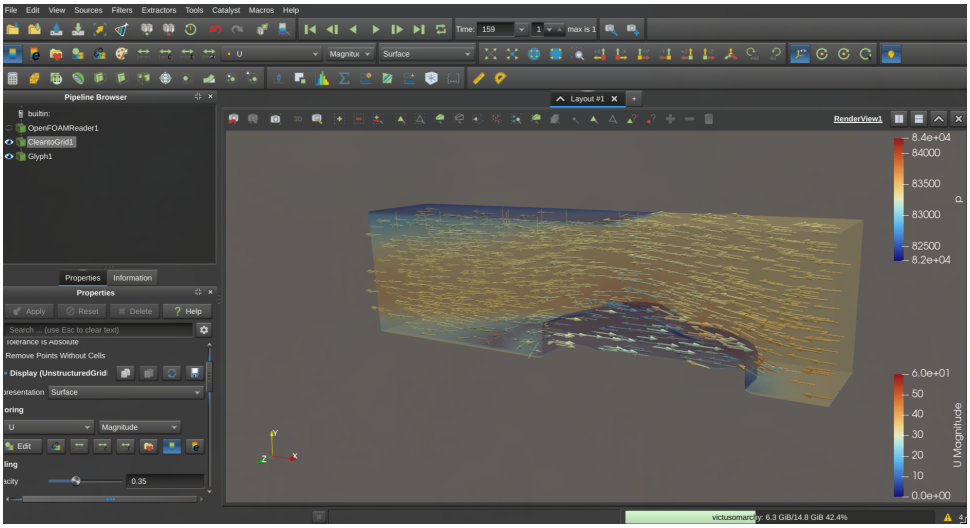
Figure 4: This was one of the first CFD simulations done. This is a case of 2 pipes of different diameters, imagine them to be welded. Flow was visualized when liquid goes into the pipe of smaller diameter and makes its way into the pipe of larger diameter



(a) 3D Model of the Car in FreeCAD

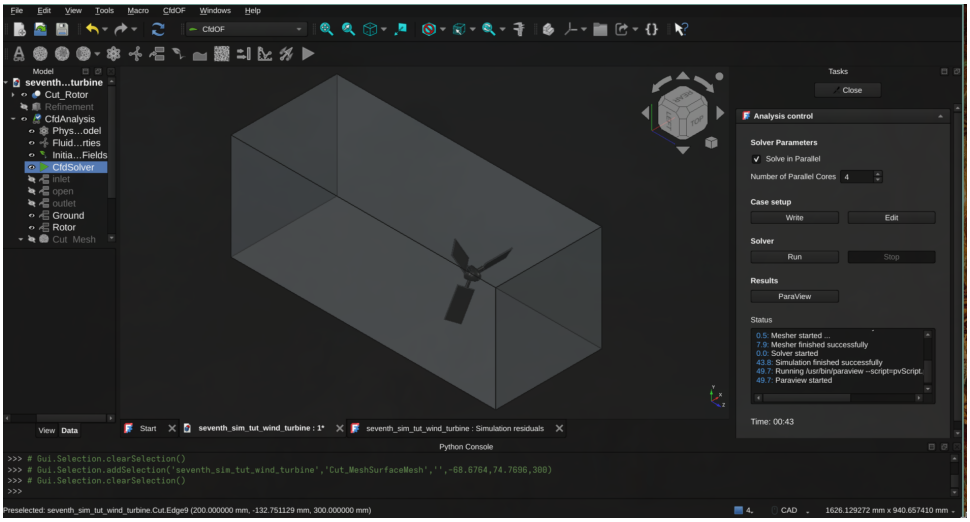


(b) Mesh Case in FreeCAD

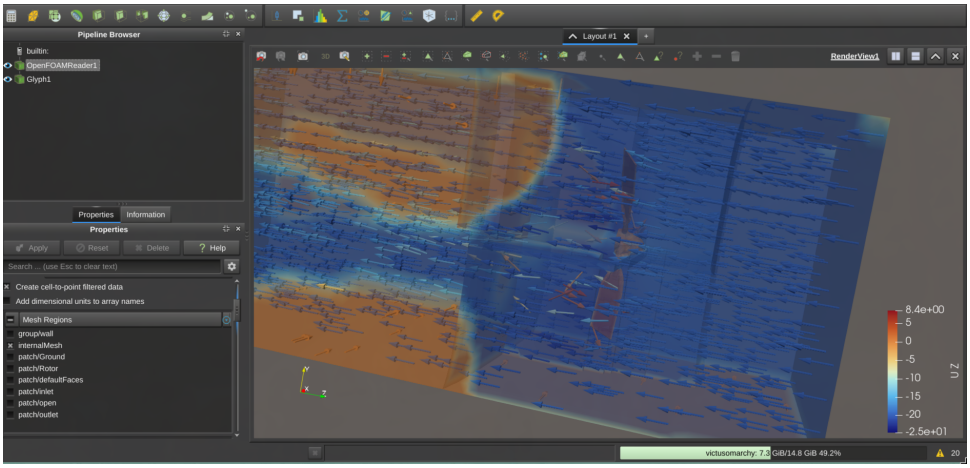


(c) Flow visualization in Paraview using Glyphs

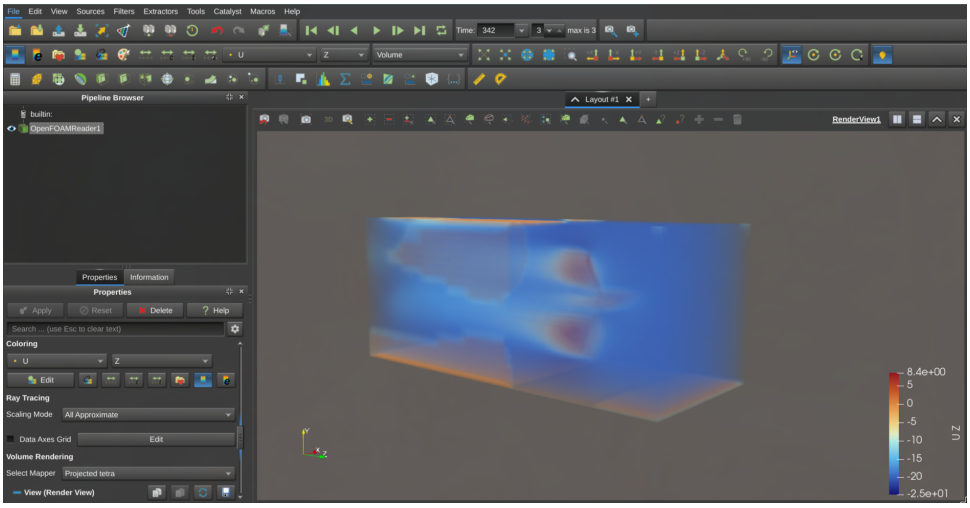
Figure 5: This case is of flow visualization around a car. A 1 to 1 scale version of a car was modelled in CAD. The technique in cases like these is to have a massive virtual wind tunnel with a length of 5-10 chords (in this case, chord being the car's length) and a sufficient width that captures a bit more than the width of the car. The car is then 'subtracted' from the wind tunnel with the help of a boolean cut function.



(a) 3D Model of the Wind Turbine in FreeCAD



(b) Flow visualization in Paraview using Glyphs

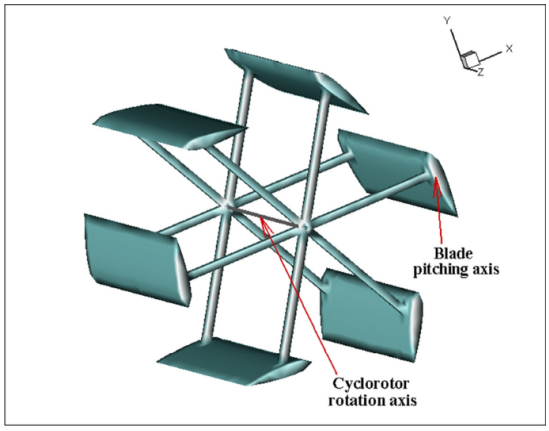


(c) Flow visualization in Paraview using a 'volume' setting.

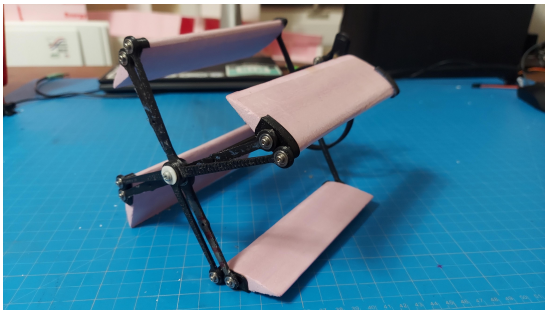
Figure 6: This case is of flow visualization around a Wind Turbine. The simulation philosophy remains the same as in the previous case.

2.2 Picking out the Cyclorotor Blade

Cylorotors traditionally have a symmetrical airfoil blades like NACA-0012, -0015, etc. The reason for this is that these blades continuously change their angle of attack and flip the direction of lift as they revolve around their rotatry axis. In such a case, it is important to have predictible lift in both directions and this allows for a stable center of pressure which helps in minimising any unwanted twists.



(a) Credit- https://link.springer.com/article/10.1007/s11012-021-01413-4



(b) Credit- Jude Schaur

Figure 7: Symmetrical Airfoils used in Cyclorotors

Symmetrical airfoils have proven to be ineffcient at low reynolds numbers (ADD REFERENCE) unlike flat plates so it makes sense to have a cyclorotor with flat plates. But flat plates are ineffcient at high reynolds number where airfoils shine. Thinking on this, the solution to such a problem could be to have a *morphing* blade. A really thin shape at the start as its taking off and then it thickness would change as per wind conditions.



Figure 8: Morphing Airfoil. Credit- The Native Lab (Rain check on this credit)

This was just a thought and incredibly complex solution which would require an incredible amount of material science and engineering. For simplicity, *Stage 1* and *Stage 2* will include flat plate simulations and physical experiments.

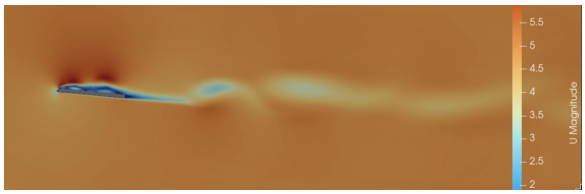
2.3 Early Simulations

The following simulations were done to test different wind tunnel sizes, refinement box sizes, different angles of attack, with and without rounded edges. For some simulations, steady state solvers were used and for others, transient solvers.

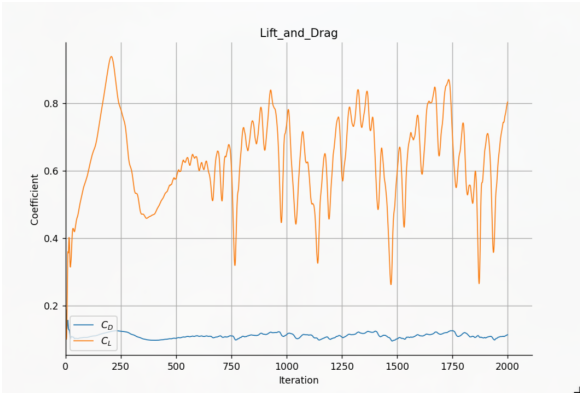
Steady-State- This solver assumes the flow field doesn't change with time- one is solving for a single, time-invariant solution where all the residuals converge to a fixed point. In terms of OpenFOAM, the force coefficients are plotted against the number of iterations. These iterations are not time-steps, it's a step (a bunch of calculations) sort of marching towards convergence.

Transient- This is a time-varient solution. It's the analysis of the entire flow- when the wind hits the surface of the blade, flow separation. vortex shedding, etc.

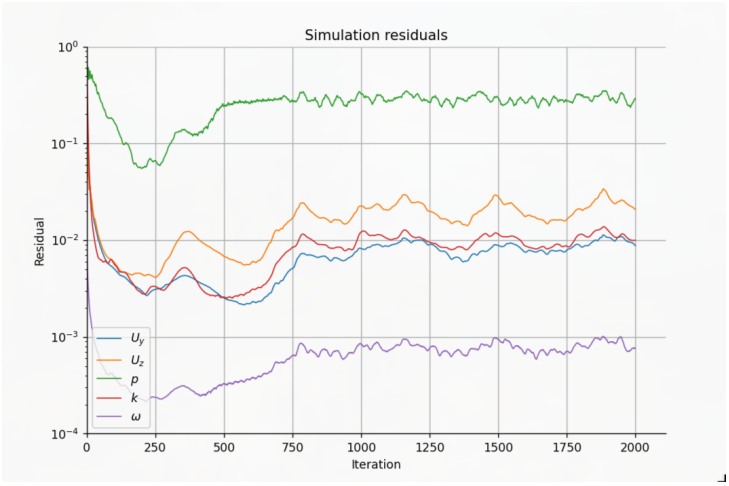
2.3.1 Steady-State, Angle of Attack = 5 degrees, Sharp Edges



(a) Flow visualization in Paraview



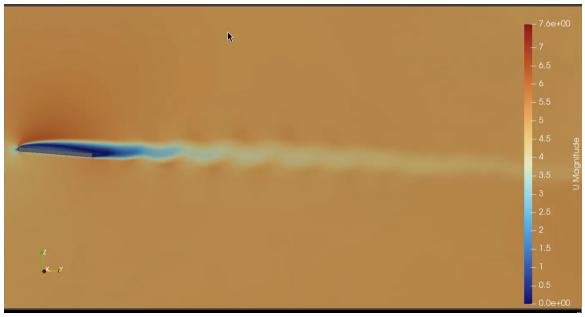
(b) Lift and Drag Coefficients



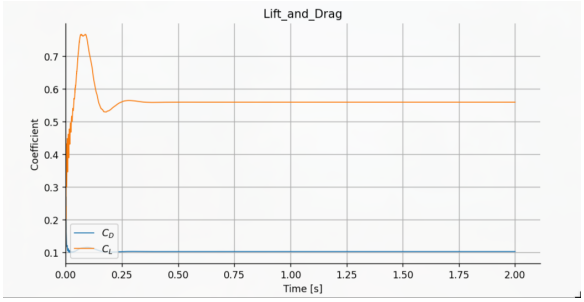
(c) Velocity, Pressure and Parameter Residuals

Figure 9: The velocity field confirms why a clear alternating wake (vortex-street-like structure) is visible downstream of the plate, meaning the underlying physics is inherently unsteady and cannot be captured by a steady solver; this run was treated as a diagnostic/troubleshooting case.

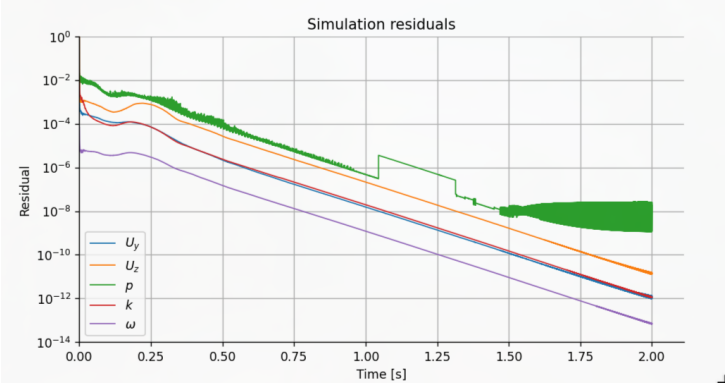
2.3.2 Transient, Angle of Attack = 5 degrees, Sharp Edges



(a) Flow visualization in Paraview. This image is a single frame from a video of the simulation.



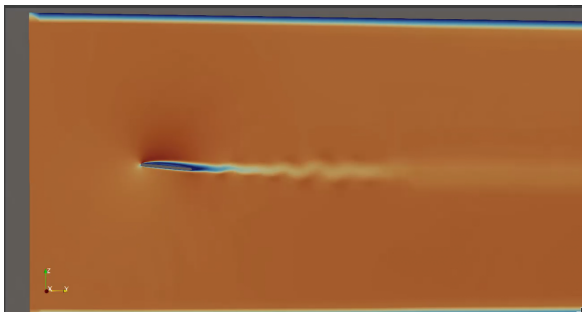
(b) Lift and Drag Coefficients



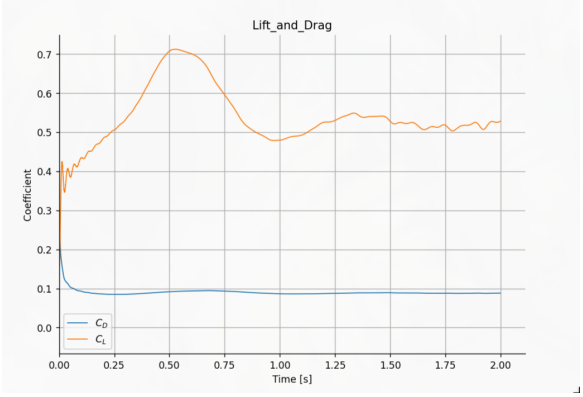
(c) Velocity, Pressure and Parameter Residuals

Figure 10: The sharp-edged plate reaches a genuinely converged steady state. After an initial transient overshoot (C_L peaking near 0.77 around $t \sim 0.1$ s), both C_L and C_D settle to constant values ($C_L \approx 0.56$, $C_D \approx 0.10$) by $t \sim 0.3$ s and remain flat for the rest of the run. Residuals decay monotonically over $\sim$10 orders of magnitude, confirming this is a fully attached, non-shedding solution.

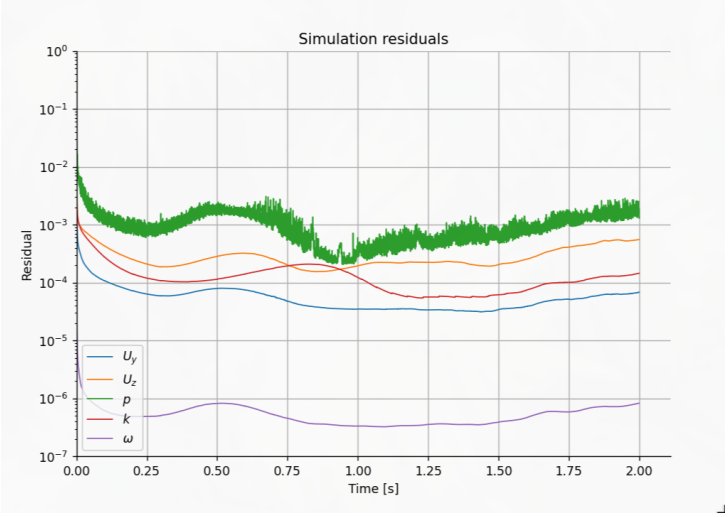
2.3.3 Transient, Angle of Attack = 5 degrees, Rounded Edges



(a) Flow visualization in Paraview. This image is a single frame from a video of the simulation.



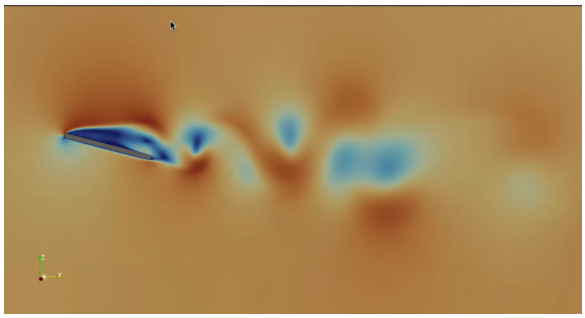
(b) Lift and Drag Coefficients



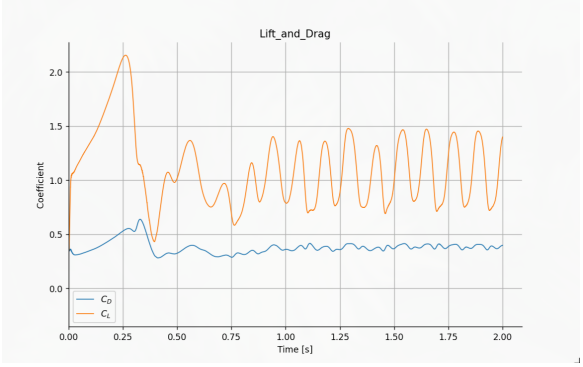
(c) Velocity, Pressure and Parameter Residuals

Figure 11: Shows the lift overshoot relaxing into a small-amplitude oscillation about $C_L \approx 0.52$–0.55, with C_D settled near 0.09 — a quasi-steady state rather than the still-rising trend seen at $t = 1$ s. This is the more trustworthy rounded-edge result at $\alpha = 5°$, though the residual plateau ($\sim 10^{-3}$) indicates mild persistent unsteadiness rather than full convergence.

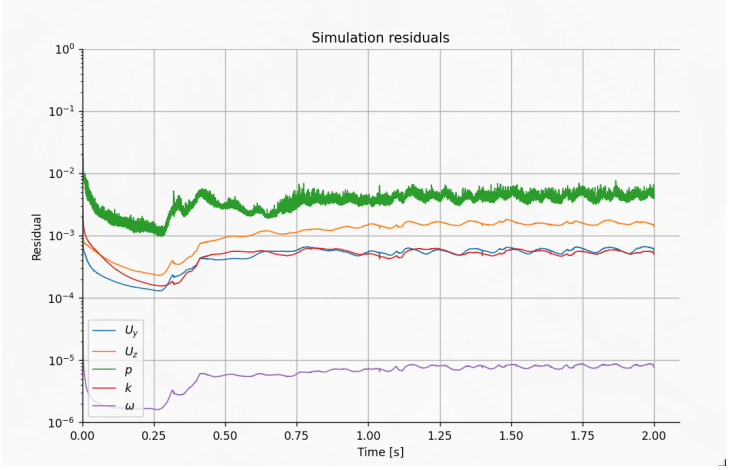
2.3.4 Transient, Angle of Attack = 15 degrees, Sharp Edges



(a) Flow visualization in Paraview. This image is a single frame from a video of the simulation.



(b) Lift and Drag Coefficients



(c) Velocity, Pressure and Parameter Residuals

Figure 12: At $\alpha = 15°$, the flow is clearly unsteady: following a strong startup-vortex spike ($C_L \approx 2.15$ near $t \approx 0.25$ s), the solution settles into a sustained periodic vortex-shedding limit cycle by $t \approx 1$–2 s, with C_L oscillating between roughly 0.7–1.5 (mean ≈ 1.1) and C_D between ~ 0.3–0.4. The residuals plateau rather than decay. Judging by the graph, this seems to be post-stall behaviour.

An important note: Most of the simulations (even the ones that were unfortunately undocumented) run for fluid flow time = 2s. This was due to disk space constraints at the time. To reduce space usage, time steps had to be reduced.

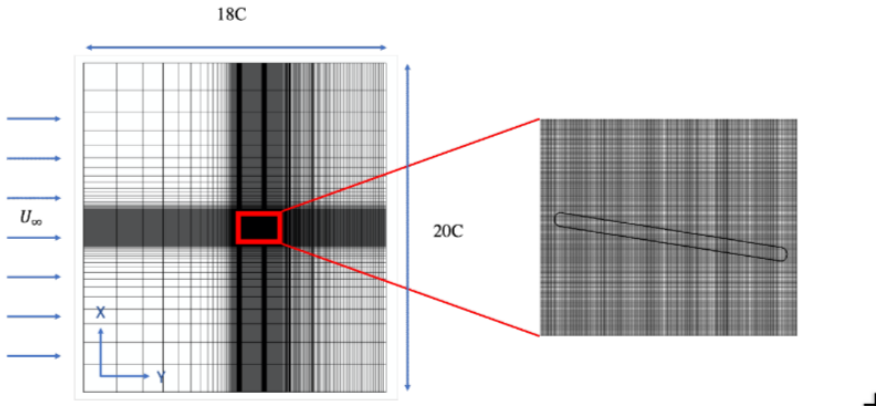
One thing about CFD simulations is that comparing 2 graphs is tricky as the simulation itself is dependent on a LOT of parameters- the size of the refinement box (relating to the "Grid Independece" test which will be explained further), Courant number (which defines how many mesh cells the fluid skips over), time steps (which affect the 'frames' one has in their simulation- having more time steps helps in visualizing better), the k and omega paramters in the solvers, etc. It is essential to have a basic understanding of how these parameters change results. Changing parameters and re-solving the cases also take up a lot of time.

2.4 Actual Simulation and Benchmarking

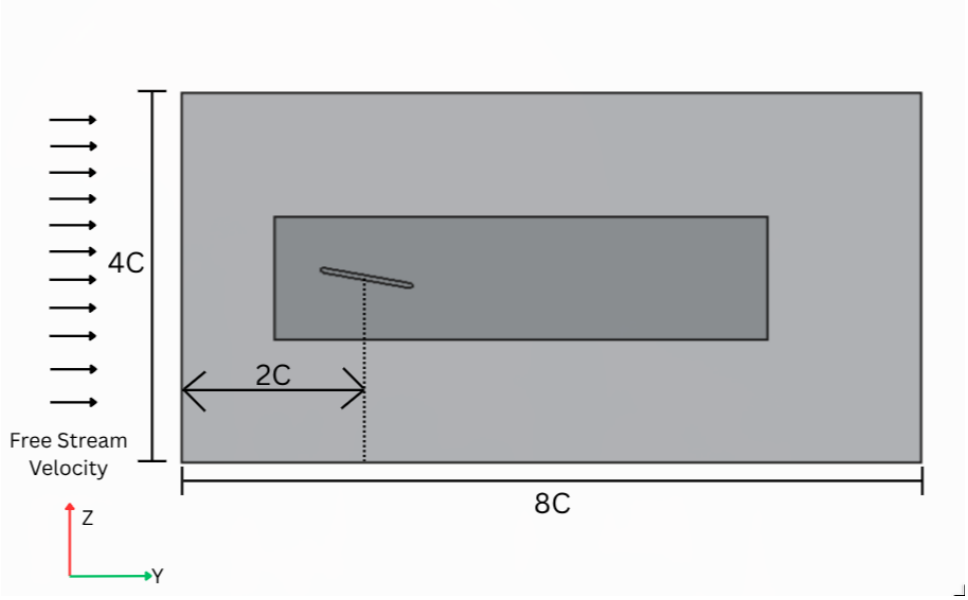
After those initial tests, it was time perform a simulation which would be benchmarked with the help of existing literature. Result validation is important before work on actual wind tunnel testing starts. Through extensive literature suyrvey, I noticed that there is sufficient literature for flat plate analysis below Re number 1000 and Re number greater than 15k. There was no little-to-no research in the 1k-10k Re number range (for flat plates) barring one key paper- Hansong Liu's thesis on airfoil aerodynamics.

This thesis was on aerodynamic characteristics of airfoils at low Reynolds numbers (add reference). It also included a section on flat plate dynamics. Liu's flow domain is different but he had used a non-dimensional quantity to plot the force coefficient values against. This non-dimensional quantity is called *Convective Time*. It is given by the following equation:

$$\tau = \frac{tU}{c} \tag{1}$$



(a) Liu's Flow Domain



(b) Flow Domain in my Simulation

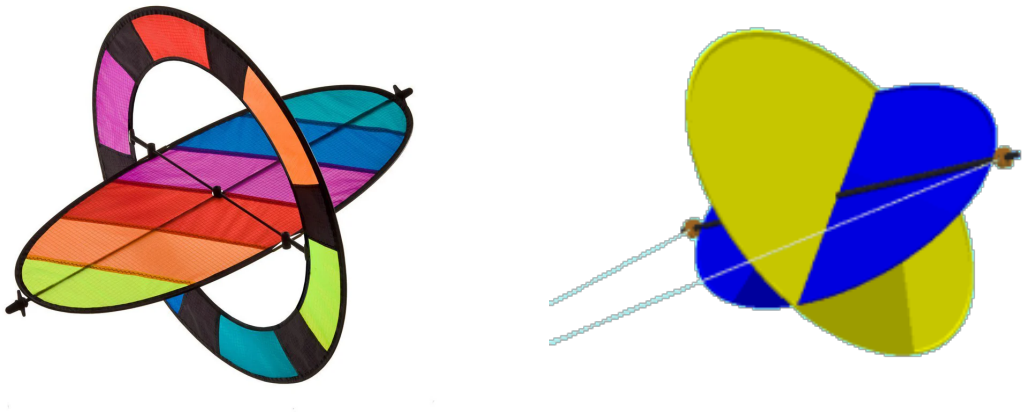
Using this, I was able to compare flow development. At the time, my simulation ran at 5000 Re and Liu's ran at 1000 Re. Not a very fair comparision since viscous effects are relatively stronger at lower

Reynolds numbers and the fluid flow tends to stay more attached before separating. <write about frequency comparison> <attach comparison graph>

2.5 Building Physical Stuff! The Rotary Kite

Along with theory, simulations and math, it is important to have hands-on experience. This also helps in enhancing physical intuitions, new questions arise with a motivation to solve them.

I started off with building the *'UFO'* rotary kite, also called the *'Rola-Rola'*. It is shown in the images below:



(a) (b)

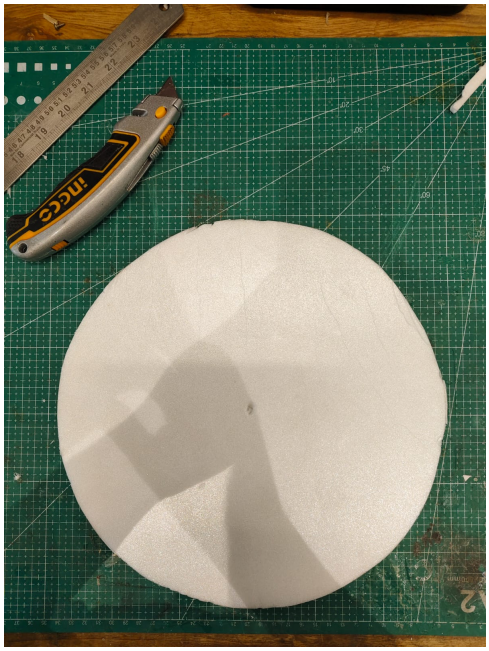
Figure 14: These type of kites rotate along the axis of the horizontal plate. The vertical plate is called the 'Ear'

The way these kites fly is very intriguing, the horizontal plate is the one that rotates, the vertical plate provides vertical stabilization. As to how they fly, this could be explained by the *Magnus Effect*- the rotating horizontal plate acts as a rotating cylinder in free wind leading to a pressure differential which results in lift. The circular/elliptical ear may add to the effect on top of working as a stabilizer.

Professor Andy Ruina had raised the question on its orientation- why does the kite stay horizontal and not vertical. If it were to rotate freely in an orientation perpendicular to its natural orientation, it would lift itself, or thrust itself sideways and not upwards. I believe it has to do with stability and how to the strings are attached on the kite. The gyro effect makes the kite stable in the orientation it stays in, but why? That's a question which require a much deeper understanding of Fluid Mechanics and a lot of experimentation.

Here were 2 iterations that I worked on:

2.5.1 Iteration 1



(a) The 'Ear'- dia of 30 cm

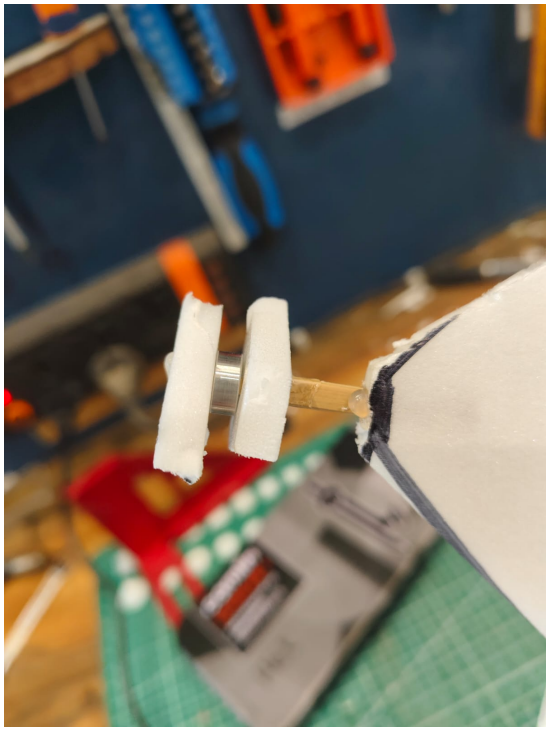


(b) Rotating Plate- 50 x 12.5 cm

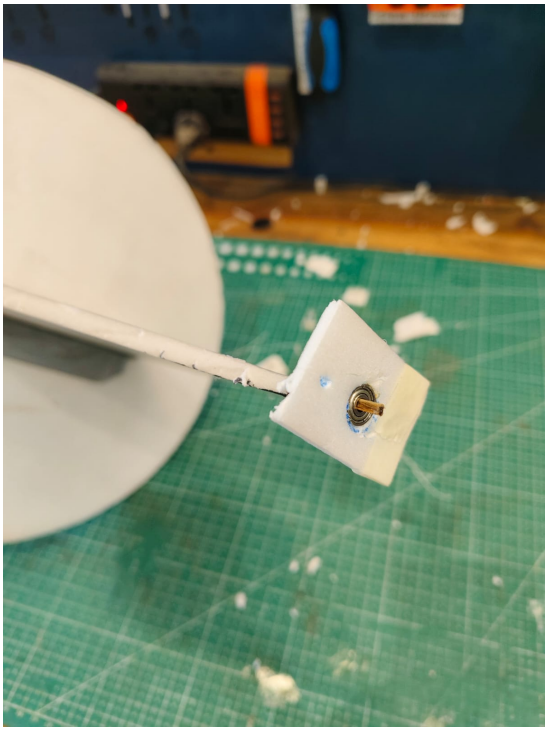


(c) Final Kite

Figure 15: The meaterial used for this kite was 6 mm*depron foam*. A rectangular hole was cut in the circular plate for placing the rotating plate. The edges of the horizontal plate were cut off to give it a smooth, triangle like shape. 2 wooden sticks were attached on the end on which bearings were friction fit.



(a) In my first design, I wrapped the string around the bearing with 2 small pieces of depron glued to the sides of the bearing to prevent the string from falling off. This did not prove to be a good design and there were issues with the string and how it was wrapped.

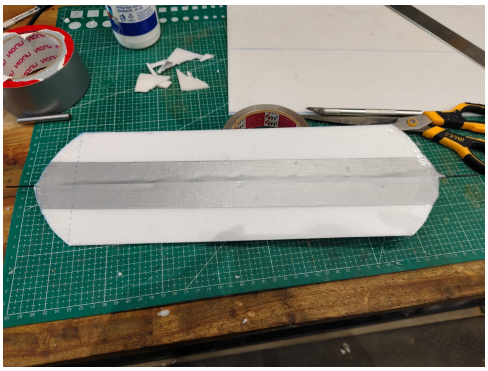


(b) In this modified version, I attached a piece of depron on the OD of the bearing and a pinned a small hole on this piece around which the string was attached. The idea was for the kite to rotate about an axis passing through the center of the bearings. But, the bearings used had too much friction and they did not have the "needing next-to-nothing to turn" factor, as a result this iteration did not fly at all.

2.5.2 Iteration 2

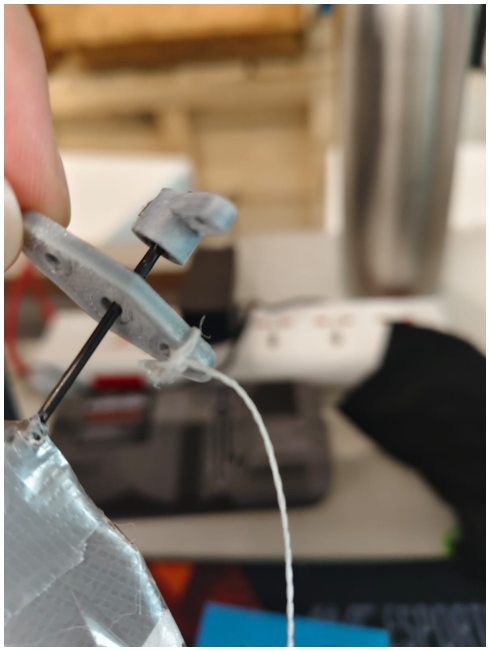


(a)

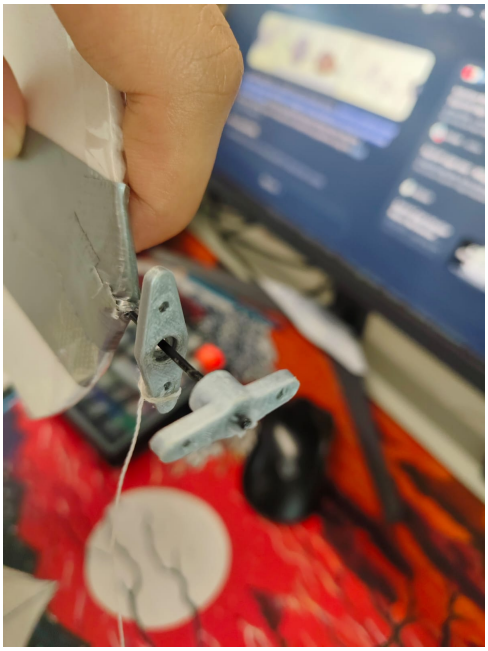


(b)

Figure 17: In my second iteration, the dimensions were reduced and 3 mm depron foam was used.These images show the rotating plate. A 2 mm carbon fibre spar was glued along the axis of rotation, providing more structural stability.



(a)



(b)

Figure 18: To attach the strings, I had made use of 3D-printed servo motor arm connectors which I had lying around. The centerbore of the arm connectors was big enough to allow some play and not be completely fit onto the shaft. This allowed for smooth rotation. In one of my discussions with Professor Andy Ruina, he pointed that having this 'gap' makes for a pretty good bearing. The magic of fluid dynamics is what takes care of this (ADD TO THIS POINT ABOUT THE 'MAGIC').

This iteration was a significant improvement over the last one and flew flawlessly. The photos shown below were of its very first flight in wind speeds of 17 kmph.



(a) Take-off



(b) Stable Flight in the air. (ESTIMATE HEIGHT BASED ON THE FLOODLIGHTS IN THE FIELD)



(c) A near bird-strike in its very first flight!

3 Stage 2

3.1 The Wind Tunnel Experiment and Design Philosophy

Once the simulation work was completed, (REMINDER: ADD THE 1K Re COMPARISON) it was time to move on to Wind Tunnel testing. The tunnel is 4 feet in length with a 2x2 feet airflow cross-section. (ADD DETAILED DESCRIPTION LINK HERE WHICH WILL TAKE THE VIEWER TO THE PART WHERE I SHOW THE ACTUAL EXPERIMENT STUFF).

Upon discussion with my supervisors, we decided to have a dual-configuration for the experiment. The flat plate would be fixed at a particular angle of attack and would also be able to freely rotate.For the first case, lift, drag and moment values will be obtained at angles of attack from 0 to 90 degrees with 5 degree increments. Same for the second case, but the takeaway from that experiment is to learn how rotating flat plates generates lift as seen in the *UFO rotary kite.*

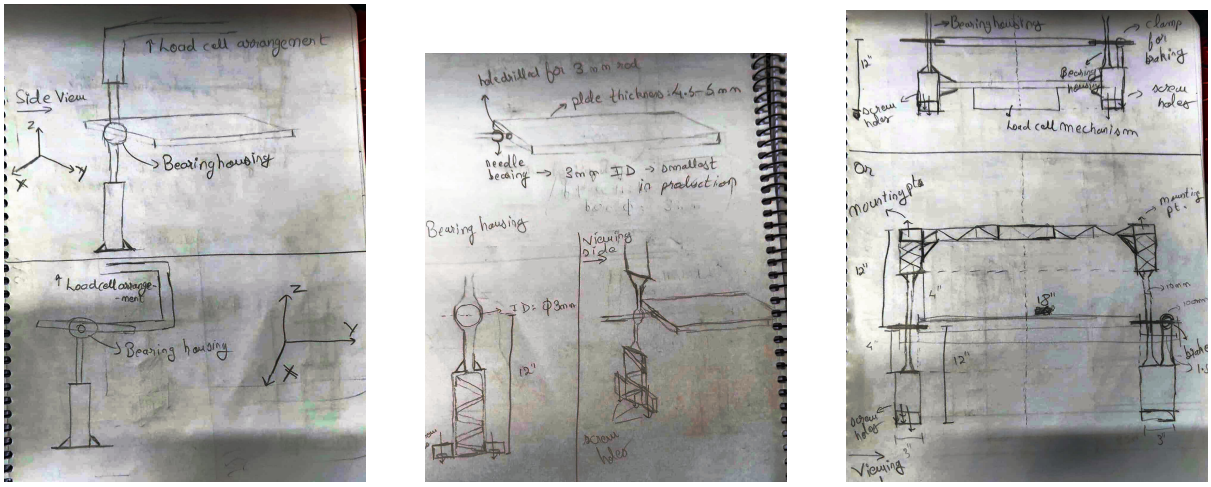
The plate would be side-mounted in order to allow rotation. All sketches to be shown in subsequent sections will be based on side-mounting.

The 'Design Philosophy' part of this section is written after realising how incredibly flawed certain designs were which I will mention in the subsequent sections. As someone who's not very fluent in structural engineering- I believe this to be a good practice in terms of 'learning from your mistakes'. Sometimes, subtle but major problems are only spotted after spending considerable time trying to design something.

3.1.1 Initial Rough Sketches with some CAD Models

This was the ideation stage of Stage 2. Below are some of rough sketches and CAD models of some ideas that were sketched out. PS: These sketches were enhanced using filters since they were sketched very lightly leading to low visibility.

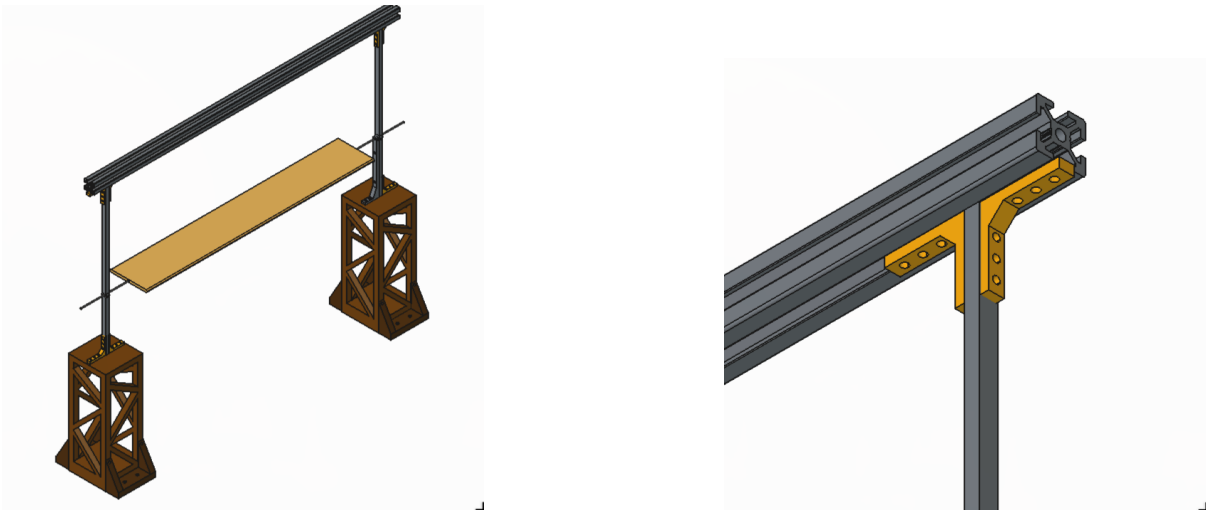
3.1.2 Sketch 1



(a) (b) (c)

Figure 20: The idea was to have a minimal visual distraction from the side mounts since flow visualization was to be seen from one of the side panes of the wind tunnel. The wind tunnel was designed to have mounting holes on its lower surface, so that's where the base mounts go. At this point in time, I was unaware of the exact location of the mounting holes, so the load cell arrangement was to be placed on the upper arms in the images. Figure c also has a configuration with the load cell arrangement placed beneath the plate on the lower surface. In the bottom section in Figure c - the complicated truss design on the top was a result of overthinking the design, but a simple aluminium extrusion was a better option.

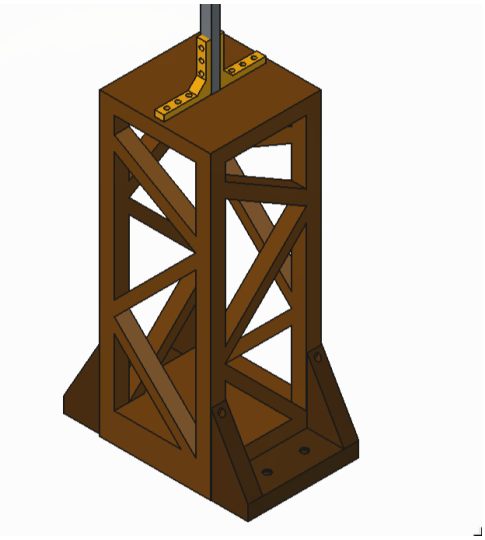
CAD Model:



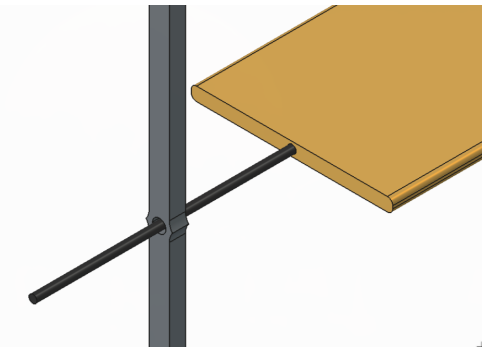
(a) Isometric view of the first design.

(b) Custom triangular braces to connect perpendicular pieces

Figure 21



(a) Base Mount



(b) Mounting arm with the bearing housing. The bearing used would be a 5 mm OD needle bearing.

Figure 22

The major flaw with a setup like this is having both base mounts and upper mounts- the forces will be distributed to both regions leading to inadequate loads to the load cells.

3.1.3 Sketch 2

This was a design used in an *Autodesk Instructables Project*:

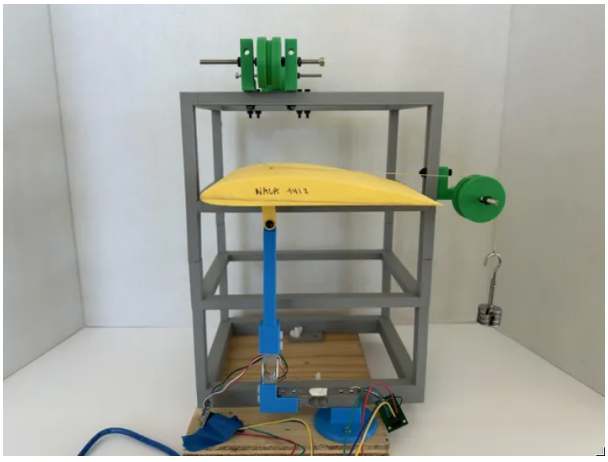


Figure 23: This uses 2 load cells the vertical one to measure drag and the horizontal one to measure lift in an 'L' arrangenment along the 'load arm'

This is how the above arrangement look for my setup:

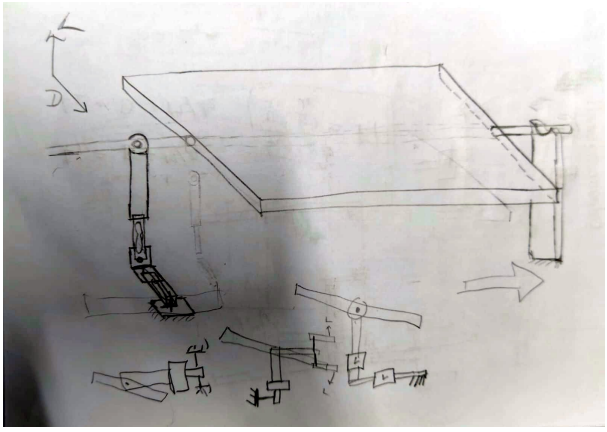
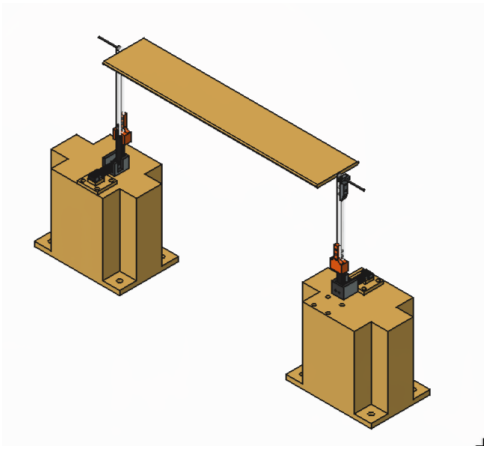
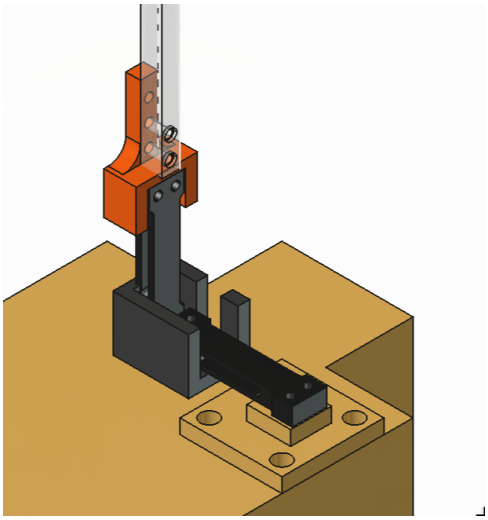


Figure 24

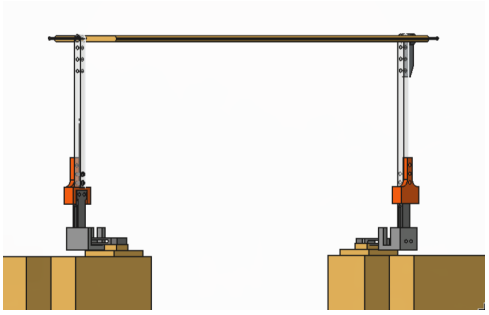
CAD Model:



(a) Complete Structure



(b) Close-up on the load cells



(c) Load cell arrangement

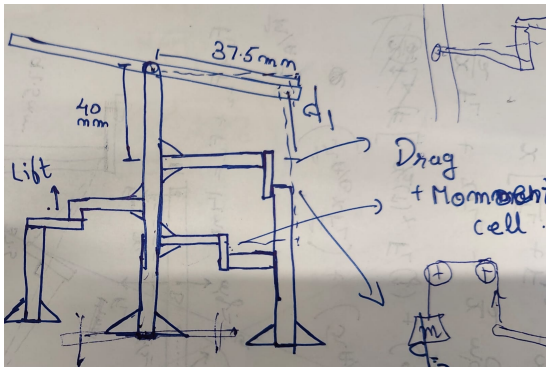
Figure 25

The primary issue with a structure like this is with the length of the arm. The 'L-shape' induces an unwanted bending moment from the structure itself which can be reduced by reducing the length of the arm. Calculations for maximum expected lift, drag and moment would decide the dimensions.

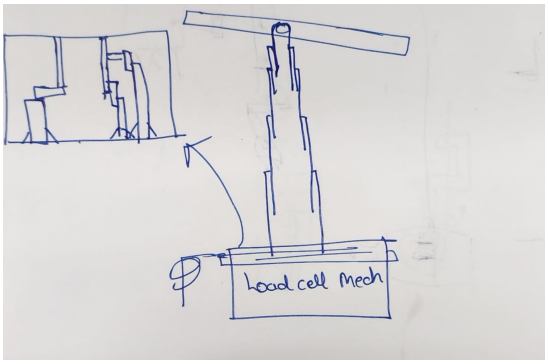
3.2 Iterations on the final design

3.2.1 Iteration 1

The first iteration involves the use of 3 load cells, 1 for lift and 2 for drag. Moment will be calculated with the help of all 3 load cells. Here are a couple of rough sketches:



(a) This was the inital idea, it was to have horizontal beams connecting to the load cells which were mounted rigid structures with triangular braces which help increase the area at the base, 'Area Moment of Inertia' increases, thereby decreasing the bending moment of the structure and increasing stability.



(b) Instead of keeping the load cells exposed, they can be kept enclosed in a box underneath the base. This way, the forces can be amplified. But, mathematical analysis on error propagation tells a completely different story on the amplification.

Figure 26: The problem with the deisgn in the first image is the "wind" and the "flow" issue. The wind will disturb the readings from the load cells. While this problem can be countered with the help of panels or enclosing the load cells in a box. The box itself will disturb the flow around the flat plate.

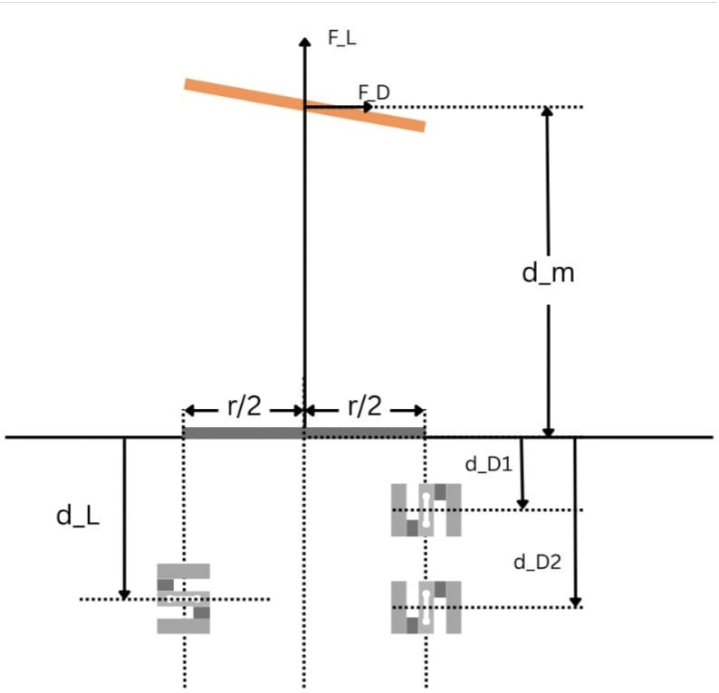


Figure 27: Schematic for Iteration 1. F_L is the lift theoretical lift force based on the free stream velocity and span of the wing. d_m is the length of the mounting arm (which will be a light-weight telescopic structure). r is the length of the base plate. d_L, d_{D1} and d_{D2} are the distances from the base plate to the load cells, more particularly the center line of the shear load cells. Although not mentioned in this diagram, but the expected load on the lift load cell is N_L and on drag load cells are N_{D1} and N_{D2}. W is the cummulative weight of the structure + the wing

Below are the calculations for the force and moment functions, expected load on the load cells and the error propagation:

Force and moment definitions. The forces and moment acting on the plate are defined in the standard aerodynamic convention, using the dynamic pressure of the freestream:

$$F_L = \frac{1}{2}\rho U^2 A\, C_L \tag{2}$$

$$F_D = \frac{1}{2}\rho U^2 A\, C_D \tag{3}$$

$$M = \frac{1}{2}\rho U^2 Ac\, C_M \tag{4}$$

where ρ is the density of the working fluid (air), A is the planform area of the plate, U is the freestream velocity, c is the chord length, and C_L, C_D, C_M are the non-dimensional coefficients of lift, drag, and pitching moment respectively. $\rho = 1.225$ kg/m^3 is taken as the standard sea-level density of air, a reasonable assumption for indoor, room-temperature wind tunnel testing where no altitude or temperature correction is needed. The planform area is $A = c \times s$, where $c = 75$ mm is fixed by the model geometry and $s = 381$ mm is the span.

Substituting these values into Equations (2)–(4) (lengths in millimetres):

$$F_L = 45.9375 \times 10^{-6}\, U^2 s\, C_L \tag{5}$$
$$F_D = 45.9375 \times 10^{-6}\, U^2 s\, C_D \tag{6}$$
$$M = 3.4453 \times 10^{-6}\, U^2 s\, C_M \tag{7}$$

with F_L, F_D in newtons, M in newton-metres, U in m/s, and s in mm.

Force balance. The floating assembly (plate, mounting arm, and load-cell attachment points) is treated as a free body under the aerodynamic loads F_L, F_D, M, its own weight W, and the reaction forces from the three load cells: one lift cell reading N_L, and two drag cells reading N_{D1} and N_{D2}.

$$N_L\,\hat{j} + F_L\,\hat{j} - W\,\hat{j} - N_{D1}\,\hat{i} - N_{D2}\,\hat{i} + F_D\,\hat{i} = \vec{0} \tag{8}$$

Dotting Equation (8) with $\hat{i}$ and $\hat{j}$ separately gives the two scalar equations:

$$N_{D1} + N_{D2} = F_D \tag{9}$$
$$N_L = W - F_L \tag{10}$$

Moment balance. Taking moments about the origin O (the pivot/shaft axis), with the lift cell offset by $r/2$ from the centreline and the two drag cells positioned at $d_m + d_{D1}$ and $d_m + d_{D2}$ from the pivot, and collecting terms along $\hat{k}$ (cross-product steps omitted):

$$-\frac{r}{2}N_L - (d_m + d_{D1})N_{D1} - (d_m + d_{D2})N_{D2} + M = 0 \tag{11}$$

Data reduction. Substituting Equation (9) into Equation (11) to eliminate N_{D2}, then solving for N_{D1} and N_{D2} individually (algebra omitted), gives the reduction equations used to recover the physical loads from actual cell readings:

$$N_{D1} = \frac{(d_m + d_{D2})F_D - \frac{r}{2}N_L - M}{d_{D2} - d_{D1}} \tag{12}$$
$$N_{D2} = \frac{M + \frac{r}{2}N_L - (d_m + d_{D1})F_D}{d_{D2} - d_{D1}} \tag{13}$$

together with Equation (10) for F_L and Equation (9) for F_D. These four equations map raw load-cell readings back to F_L, F_D, and M during actual testing.

Error propagation. Assuming each load cell carries an independent reading error δN_L, δN_{D1}, δN_{D2}, the propagated errors in the recovered quantities follow directly from Equations (10)–(13):

$$\delta F_L = \delta N_L \tag{14}$$
$$\delta F_D = \delta N_{D1} + \delta N_{D2} \tag{15}$$
$$\delta M = \frac{r}{2}\delta N_L + (d_m + d_{D1})\delta N_{D1} + (d_m + d_{D2})\delta N_{D2} \tag{16}$$

with the corresponding percentage errors reported as $\delta F_L/F_L$, $\delta F_D/F_D$, and $\delta M/M$.

Numerical evaluation. The following dimensions correspond to this iteration's balance design (drag cells mounted at the base plate, far from the pivot; lift cell offset by $r/2$ from centreline):

Quantity	Value
d_m (arm length)	300 mm
d_{D1}	30 mm
d_{D2}	80 mm
r (lift cell offset)	60 mm
W (dead weight)	3.924 N
U (freestream velocity)	5 m/s
s (span)	381 mm
Assumed cell error, δ	1% of reading

Aerodynamic coefficients for a flat plate at low-to-moderate Reynolds number were taken as indicative literature values, to be replaced with finalized values once the CFD/literature benchmarking is complete:

α	C_L	C_D	C_M (about $c/2$)
10°	0.60	0.10	0.140
15°	0.75	0.16	0.161
45°	0.60	0.72	0.093
90°	0.00	1.20	0.000

Converting these to physical loads via Equations (5)–(7), then propagating a 1% error through Equations (14)–(16), gives the following result:

α	Error in F_L	Error in F_D	Error in M
10°	14.0%	110%	397%
15°	11.0%	73%	363%
45°	14.0%	28%	1050%
90°	–	23%	undefined ($\delta M \approx 44$ N mm, vs. $M = 0$)

These calculations show that even a mere 1% error at each individual load cell propagates to a 10–14% error in lift, a 23–110% error in drag, and a 360–1050% error in the pitching moment. The root cause is geometric, not a hardware quality issue: because the drag cells sit far from the pivot ($d_m + d_{D1,2} \approx 330$–$380$ mm), each cell is mostly reacting a large couple produced by the moment and by the dead weight acting off-centre, rather than the drag force itself. The true drag and moment end up as small differences between two large, nearly-cancelling cell readings – so a small absolute error in either reading swamps the actual signal being extracted. This motivated relocating the drag cells closer to the plate in the next iteration.

3.2.2 Iteration 2

A new lift load cell was introduced, while the drag load cells were shifted under the direct line of force. Addition of the new load cell did decrease the error propagation in all 3 parameters- lift, drag and moment, but there was still scope for improvment. Below are the calculations for this iteration:

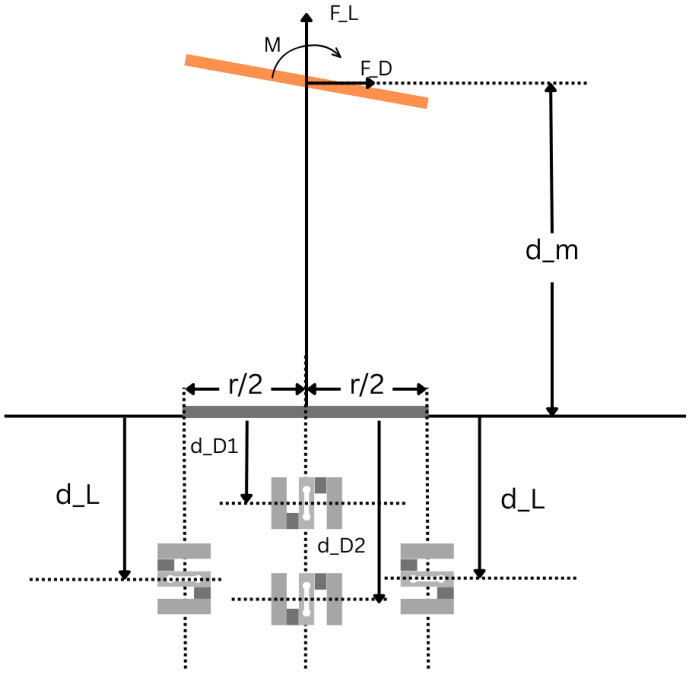


Figure 28

Setting up the free body. This geometry differs from the previous two designs in an important way: it uses *two* lift cells (at $x = -r/2$ and $x = +r/2$) in addition to the two drag cells (at $y = -d_{D1}$ and $y = -d_{D2}$, both on the centreline). A rigid body in two dimensions has only three independent equilibrium equations, but this configuration has four unknown reactions- the system is *statically indeterminate* by one degree.

Let N_{L1}, N_{L2} denote the force in the left/right lift cells (positive supporting weight), and N_1, N_2 the shear in the upper/lower drag cells (positive downstream). Using the same dimensions as Concept 1 ($d_m = 300$ mm, $d_{D1} = 30$ mm, $d_{D2} = 80$ mm, $r = 60$ mm, $W = 3.924$ N, $s = 381$ mm, $U = 5$ m/s):

$$N_{L1} + N_{L2} = W - F_L \tag{17}$$

$$N_1 + N_2 = F_D \tag{18}$$

$$M = (d_m + d_{D1})N_1 + (d_m + d_{D2})N_2 + \frac{r}{2}(N_{L2} - N_{L1}) \tag{19}$$

Three equations, four unknowns: the *sums* $N_{L1} + N_{L2}$ and $N_1 + N_2$ are recoverable, but the individual split between the two lift cells is not determinable from statics alone – it depends on the relative stiffness of each mount, which is unknown prior to fabrication.

Why the indeterminacy does not affect measurement. This is the key distinction: during an actual test, no equation needs to be *solved*- every one of the four cells returns a real, measured number. The measured readings substitute directly into Equations (17)–(19), giving exact results regardless of the indeterminacy:

$$F_L = W - (N_{L1} + N_{L2}) \tag{20}$$
$$F_D = N_1 + N_2 \tag{21}$$
$$M = (d_m + d_{D1})N_1 + (d_m + d_{D2})N_2 + \frac{r}{2}(N_{L2} - N_{L1}) \tag{22}$$

Newton's laws hold regardless of whether the supporting structure is statically determinate. The indeterminacy only becomes a problem *before* the structure is built, when trying to predict what each cell will read in order to select its capacity – since the true load split depends on relative mount stiffness, sizing must proceed from a bound rather than an exact prediction.

Sizing bound (Case A – symmetric lift cells). Assuming the two lift cells are nominally identical, the most natural bound absent better information is $N_{L1} = N_{L2}$, which sends the $\frac{r}{2}(N_{L2} - N_{L1})$ term in Equation (19) to zero. The drag pair then carries the entire moment:

$$N_1 = \frac{(d_m + d_{D2})F_D - M}{\Delta} = \frac{380F_D - M}{50} \tag{23}$$
$$N_2 = F_D - N_1 \tag{24}$$
$$N_{L1} = N_{L2} = \frac{W - F_L}{2} \tag{25}$$

α	F_L	F_D	M	$N_{L1} = N_{L2}$	N_1 / N_2
10°	0.263	0.044	4.594	1.831	$+0.241$ / -0.197
15°	0.328	0.070	5.284	1.798	$+0.426$ / -0.356
45°	0.263	0.315	3.052	1.831	$+2.333$ / -2.018
90°	0.000	0.525	0.000	1.962	$+3.991$ / -3.466

Unlike Concept 1, there is no tare at $U = 0$: because weight splits symmetrically across two lift cells with zero net moment, the $\pm\frac{r}{2}W/\Delta$ dead-weight couple that contaminated Concept 1's drag cells does not appear here. This preserves the full cell range for actual aerodynamic signal.

Peak loads: drag cells reach ~ 4 N (requiring ~ 5 N capacity cells), while each lift cell reaches only ~ 2 N – half of what Concept 1's single lift cell needed, since the load is now shared between two cells.

Error propagation. Taking 0.05 %FS cell error gives $e_D = 2.5$ mN for each drag cell and $e_L = 1$ mN for each lift cell:

$$\delta F_L = e_{L1} + e_{L2} = 2 \text{ mN} = 0.002 \text{ N} \tag{26}$$
$$\delta F_D = e_1 + e_2 = 5 \text{ mN} = 0.005 \text{ N} \tag{27}$$
$$\begin{aligned}\delta M &= (d_m + d_{D1})e_1 + (d_m + d_{D2})e_2 + \frac{r}{2}(e_{L1} + e_{L2}) \\ &= 330(0.0025) + 380(0.0025) + 30(0.002) = 1.835 \text{ N·mm}\end{aligned} \tag{28}$$

α	Error in F_L	Error in F_D	Error in M
10°	0.76%	11.4%	39.9%
15°	0.61%	7.1%	34.7%
45°	0.76%	1.6%	60.1%
90°	–	1.0%	undefined ($M = 0$); 34.7% of peak M

Interpretation. The dual lift cells in this concept give the best lift accuracy of the three designs and eliminate the dead-weight tare problem seen in Concept 1. However, the drag cells remain 300+ mm from the plate – the same long-arm amplification that broke Concept 1's moment channel persists here, simply without the additional weight-couple contamination on top of it. Splitting the lift measurement into two cells improved lift accuracy and revealed roll/ left–right asymmetry information, but did not address the underlying geometric cause of poor moment accuracy: the drag cells were never relocated closer to the plate.

The static indeterminacy itself is not the limiting factor here – Equations (20)–(22) still return exact results from real measured data, independent of how the redundant structure internally distributes load. The genuine limitation is geometric, carried over directly from Concept 1: cells placed far from the plate force the moment calculation to rely on comparatively large drag-cell readings, degrading accuracy relative to Concept 2's close-mounted design.

3.2.3 Iteration 3

Keeping in mind the error amplification from the two previous cases, in this iteration, the drag load cells were shifted above the base plate with the lift load cell being shifted directly under the line of force of the plate. This way the lift load cell only measures the lift and the drag load cells collectively measure the moment.

Here is the schematic:

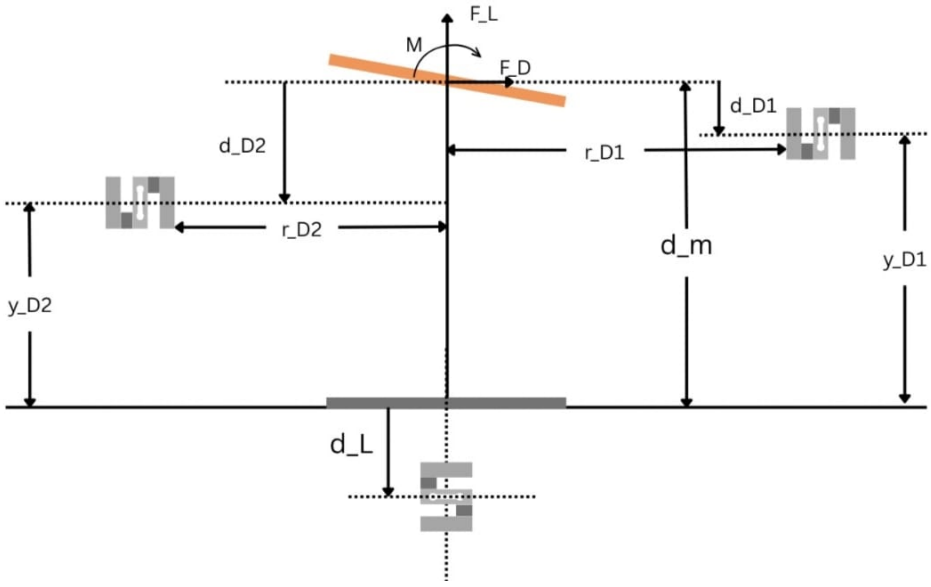


Figure 29: The additional variables are y_{D1}, y_{D2}, r_{D1} and r_{D2}. They are the vertical and horizontal mount lengths.

Below are the calculations expected load on the load cells and the error propagation:

Force balance. With the lift cell mounted directly on the pivot axis (no offset), the vertical equilibrium gives a fully decoupled lift channel from the outset:

$$N_L = W - F_L \tag{29}$$

$$N_{D1} + N_{D2} = F_D \tag{30}$$

Because there is no lateral offset between the lift cell and the plate centreline in this design, N_L never appears in the moment equation below – unlike Iteration 1, lift measurement here has zero cross-talk with the drag/moment channels by construction.

Moment balance. Taking moments about the pivot O, with the drag cells positioned at d_{D1} and d_{D2} directly from the pivot (cross-product steps omitted):

$$M = d_{D1}N_{D1} + d_{D2}N_{D2} \tag{31}$$

Data reduction. Substituting Equation (30) into Equation (31) to eliminate each unknown in turn (algebra omitted) gives the reduction equations used to recover the physical loads from actual cell readings:

$$N_{D1} = \frac{d_{D2}F_D - M}{d_{D2} - d_{D1}} \tag{32}$$

$$N_{D2} = \frac{M - d_{D1}F_D}{d_{D2} - d_{D1}} \tag{33}$$

together with Equation (29) for F_L and Equation (30) for F_D.

Error propagation. Assuming an equal reading error e on both drag cells ($\delta N_{D1} = \delta N_{D2} = e$), the error in moment follows directly from Equation (31):

$$\delta M = d_{D1}\delta N_{D1} + d_{D2}\delta N_{D2} \tag{34}$$

$$\frac{\delta M}{M} = \frac{(d_{D1} + d_{D2})\, e}{d_{D1}N_{D1} + d_{D2}N_{D2}} \tag{35}$$

with lift and drag errors following as before:

$$\delta F_L = \delta N_L \tag{36}$$

$$\delta F_D = \delta N_{D1} + \delta N_{D2} \tag{37}$$

Numerical evaluation. Dimensions for this iteration:

Quantity	Value
d_{D1}	25 mm
d_{D2}	125 mm
W (dead weight)	0.981 N
U (freestream velocity)	5 m/s
s (span)	381 mm
Assumed cell error, δ	1% of reading

Using the same indicative literature coefficients as before, converted to physical loads via Equations (5)–(7), then propagated through Equations (36)–(35):

α	Error in F_L	Error in F_D	Error in M
10°	2.7%	1.0%	1.0%
15°	2.0%	1.0%	1.0%
45°	2.7%	1.3%	5.0%
90°	–	1.5%	6.0% (of peak M)

Interpretation. Compared to Iteration 1 and 2, pulling the drag cells in close to the plate and placing the lift cell directly on the pivot axis reduces the error amplification by up to roughly two orders of magnitude across all three channels. The moment recovered in Equation (31) is now the sum of two comparably-sized, mostly-additive terms rather than the residual of two large, near-cancelling numbers – so a 1% cell error stays close to a 1% error in the final measured quantity.

3.2.4 Iteration 4

Concept 4 combines Concept 2's dual-lift-cell arrangement with Concept 3's close-to-plate drag cell placement: two lift cells at $x = -r/2$ and $x = +r/2$, and two drag cells pulled up near the plate rather than mounted at the base. As with Concept 3, four reactions against three equilibrium equations makes this system statically indeterminate by one degree.

Using dimensions consistent with Concept 2's drag cell placement and Concept 3's dual-lift geometry: $d_m = 300$ mm, $y_{D1} = 275$ mm ($d_{D1} = 25$ mm), $y_{D2} = 175$ mm ($d_{D2} = 125$ mm), $\Delta = 100$ mm, $r = 60$ mm ($r/2 = 30$ mm), $W = 0.981$ N (100 gf).

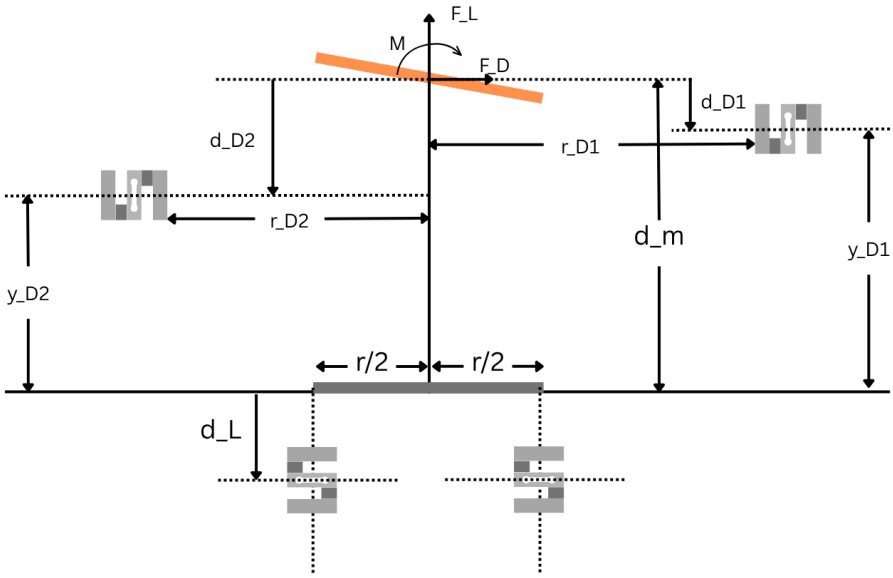


Figure 30: Schematic for Iteration 4

Force balance and Moment Balance

$$N_{L1} + N_{L2} = W - F_L \tag{38}$$

$$N_1 + N_2 = F_D \tag{39}$$

$$M = d_{D1}N_1 + d_{D2}N_2 + \frac{r}{2}(N_{L2} - N_{L1}) \tag{40}$$

As with Iteration 2, real cell readings feed directly into Equations (38)–(40) on test day and return exact results, the indeterminacy only affects predicting the split before the structure exists, since the true division of load depends on the relative stiffness of the four mounts.

Sizing bound (symmetric lift assumption). Absent better information, assuming nominally identical lift cells split the load evenly ($N_{L1} = N_{L2}$) sends the differential term in Equation (40) to zero:

$$N_{L1} = N_{L2} = \frac{W - F_L}{2} \tag{41}$$

$$N_1 = \frac{d_{D2}F_D - M}{\Delta} \tag{42}$$

$$N_2 = F_D - N_1 \tag{43}$$

α	$N_{L1} = N_{L2}$	N_1	N_2
10°	0.359 N	$+0.0091$ N	$+0.0349$ N
15°	0.326 N	$+0.0347$ N	$+0.0353$ N
45°	0.359 N	$+0.3633$ N	-0.0483 N
90°	0.491 N	$+0.6563$ N	-0.1313 N

N_1 and N_2 are identical to Iteration 3's values, since the drag-cell geometry is unchanged. The gain is on the lift side: each lift cell now peaks at 0.49 N instead of Iteration 3's single 0.98 N cell, allowing a smaller, more sensitive lift cell to be specified (e.g. 0.5 N FS instead of 1 N FS).

Reduction.

$$F_L = W - (N_{L1} + N_{L2}) \tag{44}$$

$$F_D = N_1 + N_2 \tag{45}$$

$$M = d_{D1}N_1 + d_{D2}N_2 + \frac{r}{2}(N_{L2} - N_{L1}) \tag{46}$$

Error propagation. With drag cells sized as in Iteration 3 (1 N / 0.2 N capacity) giving $e_1 = 0.5$ mN, $e_2 = 0.1$ mN at $0.05\,\%$FS, and lift cells sized at 0.5 N each giving $e_{L1} = e_{L2} = 0.25$ mN:

$$\delta F_L = e_{L1} + e_{L2} \tag{47}$$

$$\delta F_D = e_1 + e_2 \tag{48}$$

$$\begin{aligned}\delta M &= d_{D1}e_1 + d_{D2}e_2 + \frac{r}{2}(e_{L1} + e_{L2}) \\ &= 25(0.5) + 125(0.1) + 30(0.25 + 0.25) \\ &= 12.5 + 12.5 + 15 = 40 \text{ mN·mm} = 0.040 \text{ N·mm}\end{aligned} \tag{49}$$

α	δF_L	$\delta F_L/F_L$	δF_D	$\delta F_D/F_D$	$\delta M/M$
10°	0.0005 N	0.19%	0.0006 N	1.4%	0.87%
15°	0.0005 N	0.15%	0.0006 N	0.9%	0.76%
45°	0.0005 N	0.19%	0.0006 N	0.19%	1.31%
90°	–	–	0.0006 N	0.11%	0.76% (of $M_{\max}$)

Comparison across all four iterations.

	Iteration 1	Iteration 2	Iteration 3	Iteration 4
Statically determinate?	Yes	No	Yes	No
Lift error	11–14%	0.6–0.8%	2–3%	**0.15–0.19%**
Drag error	24–110%	7–11%	1-1.5%	**0.2–1.4%**
Moment error	360–1050%	35–60%	1–6%	**0.8–1.3%**
Dead-weight tare on drag	Yes	None	None	None

Interpretation. This iteration is the best-conditioned design of the four evaluated. It inherits iteration 3's well-placed drag cells (small moment arms, no large-number cancellation) while also halving the peak load on each lift cell by splitting it across two cells which additionally allows detection of spanwise or roll asymmetry, the same benefit iteration 2 offered. The moment channel performs almost as well as iteration 3's, only marginally worse because the $\frac{r}{2}(N_{L2} - N_{L1})$ differential-lift term introduces a small additional error path that iteration 3's single on-axis lift cell does not have.

The cost of this performance is the same one-degree static indeterminacy carried by Iteration 2. This does not corrupt the measurement – Equations (44)–(46) remain exact regardless of the internal load split. But it introduces practical considerations:

- The two lift cells must be nominally matched, or the split calibrated using a known off-axis dead load rather than assumed to be $50/50$.
- If the supporting frame is not perfectly rigid, the true load split can drift with temperature or preload.
- Since the drag load cells sit near the plate, nearer as in vertically and not horizontally, this would lead to design challenges, placing a box to shield the load cells from the wind under the sides of the plate will lead to flow disturbances which would lead to inaccurate readings.

Balancing the math against the physics. What emerges from comparing these four concepts is fundamentally an optimization problem, where two competing factors pull the design in opposite directions. Moving the load cells closer to the plate reduces error propagation, smaller moment arms mean the measured forces are closer to the actual signal being measured, rather than a small residual buried inside large, nearly-cancelling numbers. But moving hardware closer to the plate also increases the risk of disturbing the boundary layer and the flow field around the model itself, contaminating the very measurement the balance is meant to take cleanly. It's a tradeoff between the *Mathematics* and the *Physics* of the flow.

3.2.5 Iteration 5

Iteration 5 was about incoerporating a 4-axis load cell mechanism. All load cells would sit in a box and no load cells would be exposed as given in iterations 3 and 4. This decision was a result of the constraints in the wind tunnel. There were specific mounting holes, it didn't make sense to have load cells exposed and have their mounting structures across ends of the tunnel, the mounting holes were not spaced out throughout the base.

Here is the 3D schematic of the Balance:

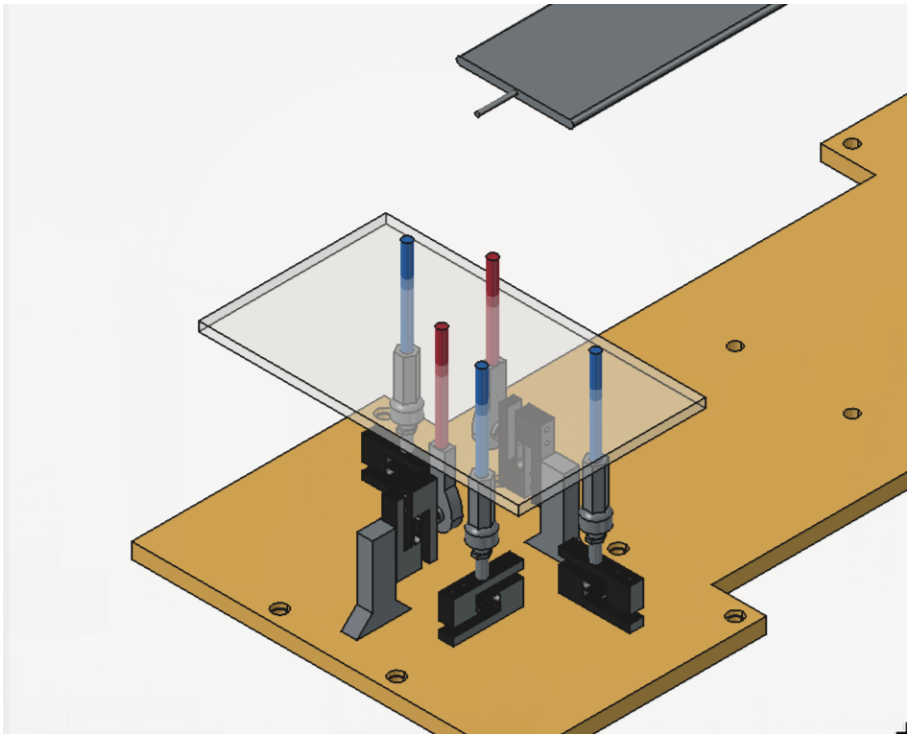
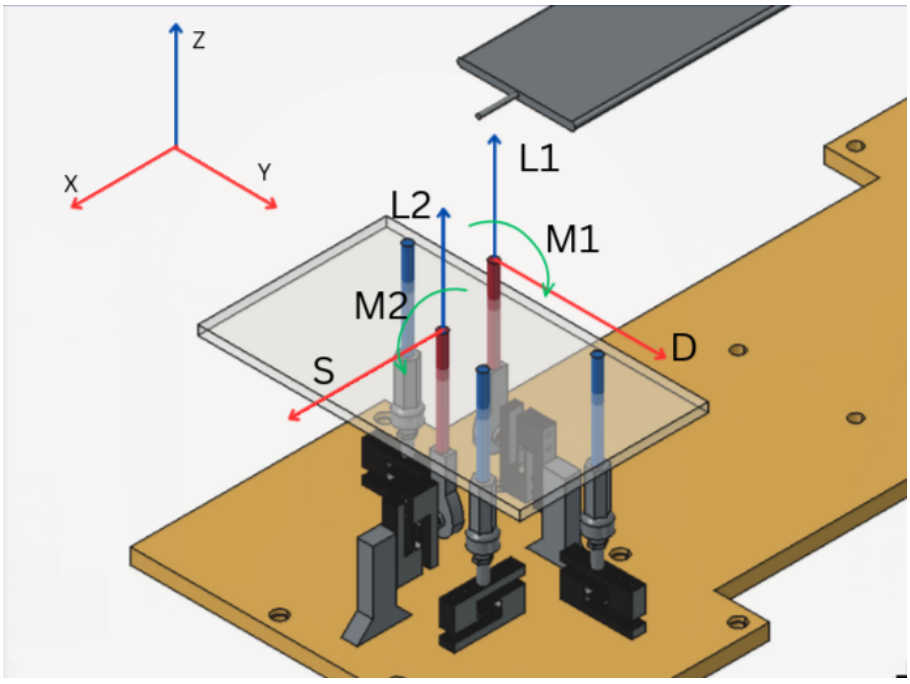


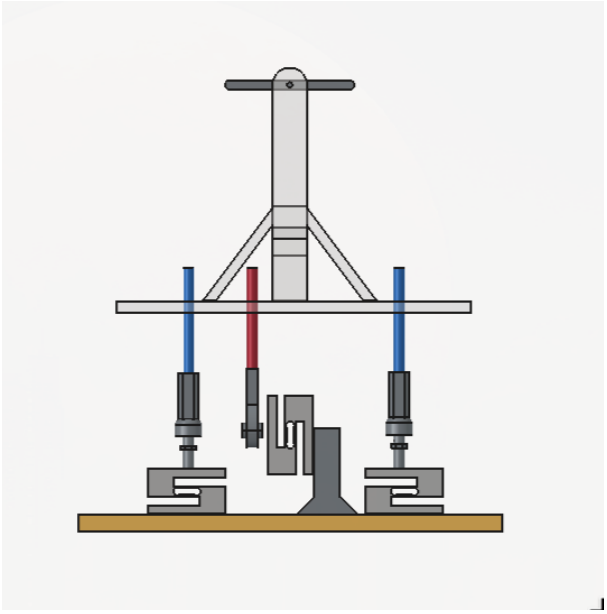
Figure 31: The yellow-ish base will be made of wood and mounted onto the mounting holes in the tunnel. Load cells will be enclosed in an acryllic box. The top grey plate is the wing. An acryllic mounting arm will connect the box and the wing. The red links are the *drag* load cells, whereas the blue links are the *lift* load cells.

Here are the forces and moments measured:

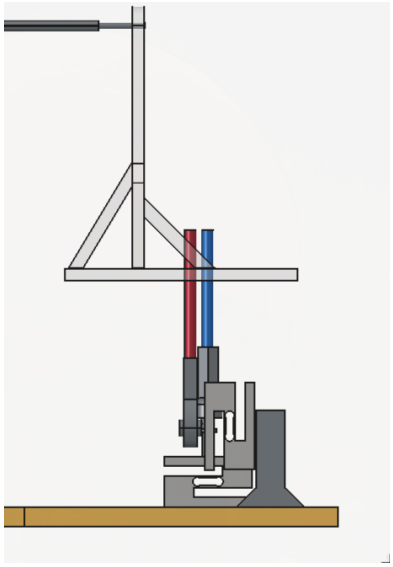


- L1 – Total lift force measured by 2 load cells in the ZY plane.
- D – Drag force measured by 1 load cell in the ZY plane.
- M1 – Moment in the ZY plane, bending moment of the wing.

- L2 – Lift force measured by one load cell in the Zx plane.
- S – The side force of the plate.
- M2 – The bending moment across the span of the plate, or the flexing of the plate.



(a) Load cell arrangement for forces and bending moment in ZY plane



(b) Load cell arrangement for forces and bending moment in ZX plane

Axial ball and socket joints will be used to connect the lift load cells and rod end bearings will be used to connect the drag load cells.

3.2.6 Iteration 6

The issue with the previous iteration is the inability of the mechanism to prevent motion of the platform in all its degrees of freedom.

A flat plate (not the wing, but the platform on which it will be mounted) is supposed to be restricted in certain ways such that we are able to take take the needed lift, Drag and/or Moment readings. Since the platform has 6 DOF (X,Y,Z,roll,pitch,yaw)- there should be at minimum 6 rods that support the structure and keep it fixed in the air.

From those 6 rods- some links will be connected to load cells to measure specific forces and moments.

It is important to completely or nearly decouple or prevent crosstalk between the load cells so as to not poison the readings.

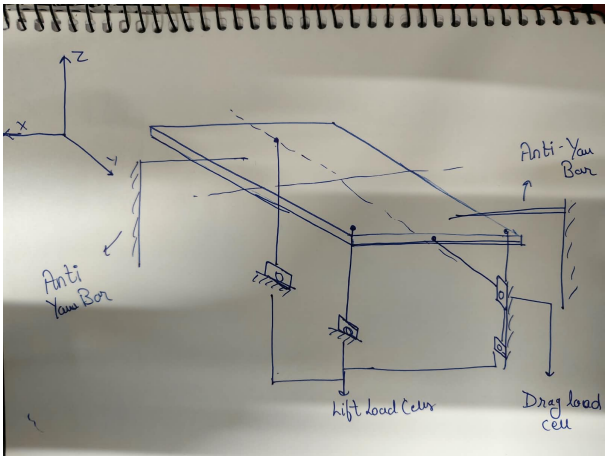
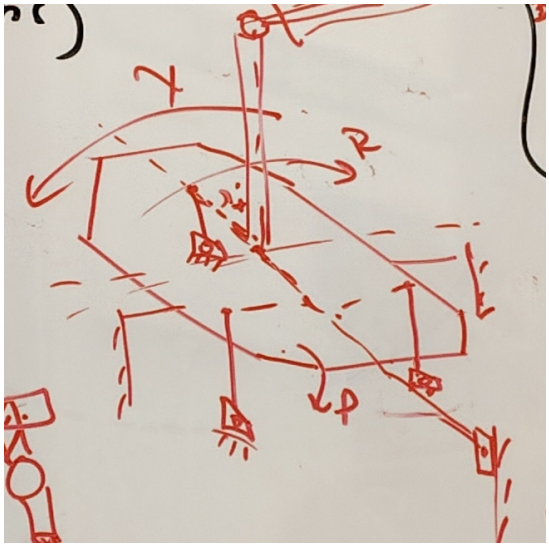


Figure 33: This design makes use of 4 load cells- 3 to measure lift and 1 for drag.

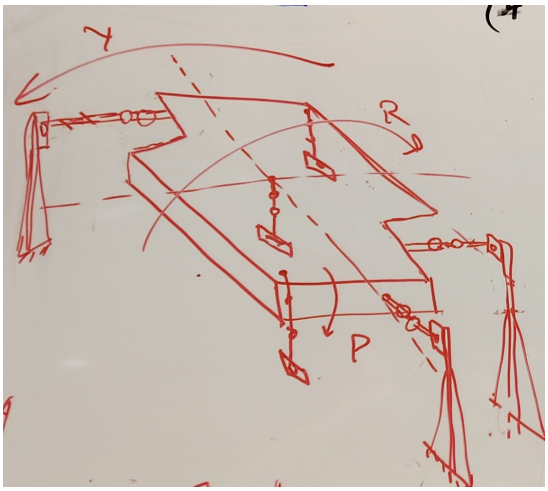
The lift load cells can also be used to measure the moment. The links connecting the lift load cells are used for measuring forces along Z-axis, same for the drag load cell in the Y-axis. These 4 links also prevent the roll and pitching of the platform. The 2 links on the sides of the platform prevent the yaw and the sideways motion along the X-axis.

The platform need not be a rectangular one and can have a more octagonal or a triangular shape (As can be seen in the next iteration). This is because a rectangular plate may tend to rotate along the axis parallel to the line joining the mount points of the lift load cell links.

3.2.7 Iteration 7



(a) Octagonal platform



(b) Different configuration of lift load cells, with load cells being used to measure yaw as well- turning it into a 6 DOF load-cell mechanism. (Y- yaw R-roll P-pitch)

3.2.8 Iteration 8

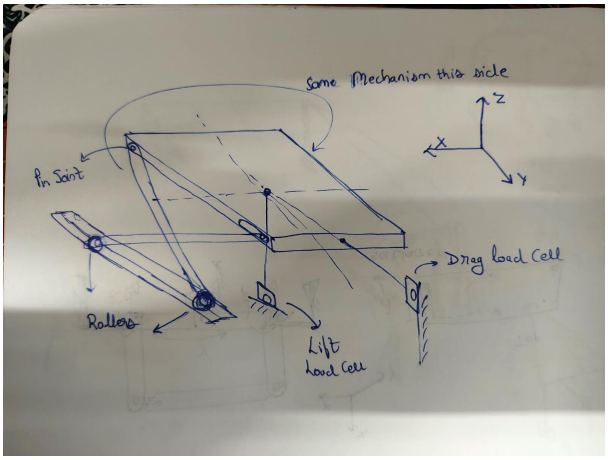


Figure 35: A 2 load cell configuration

Here is another way looking at load cell mechanisms- since in the autorotation case, we only care about lift and drag- something which only these 2 load cells are tasked to measure. A roller and slider mechanism is used to allow the motion along Z-axis (for lift) and motion along Y-axis (for drag).

3.2.9 Testing the Mounting Rig

The first physical thing that was built was the mounting rig. This one was a prototype to test the 'low-friction' capability of commercial ball bearings.

I had used 13 mm OD and 4 mm ID bearings since I had a 4 mm carbon shaft at the time. The outer cover of the bearings were taken off, put in a beaker with acetone solution. This beaker was then put in an *Ultrasonic Cleaner*. The cleaner removes all of the grease in the bearings.

Once the bearings dried, they were press fit onto laser-cut acryllic pieces.



(a) Mounting Rig



(b) I used a 5 inch by 75 mm depron piece. The depron piece was cut in half and these halves were taped onto the shaft.

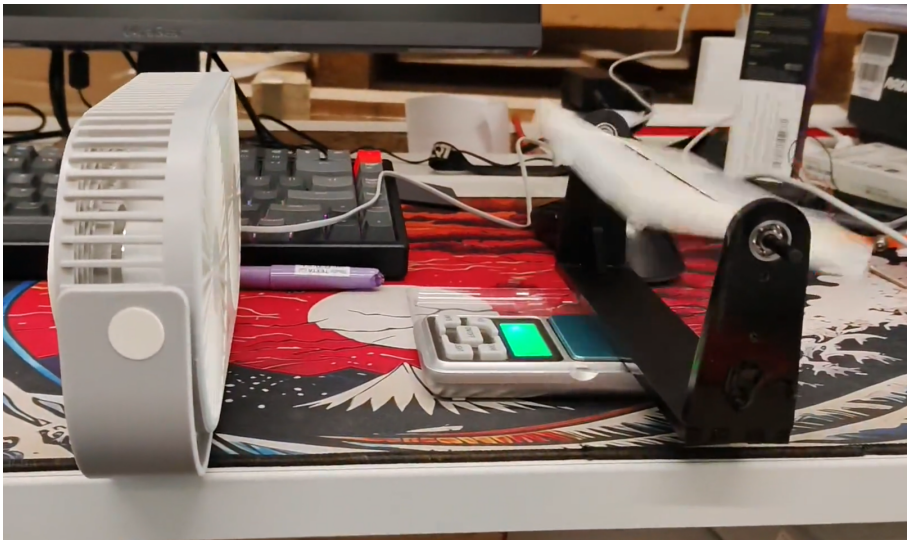


Figure 37: A table and a weighing scale was used to make a *'makeshift open wind tunnel'*.

In my tests, the plate preferred to rotate in one direction only. This was most likely due to the inconsistency in cutting the plate. It was not exactly a 50/50 cut.

Document will be updated as and when progress is made

Last Updated: November 11, 2026